

From Forecasts to Batteries: Grouped Decision-Focused Model Averaging for Capacity Commitment and Storage Siting on Five Continents

Quang-Vinh Dang

British University Vietnam, Hung Yen, Vietnam

vinh.dq4@buv.edu.vn ORCID: 0000-0002-3877-8024

Abstract

Grid operators must commit capacity a day ahead, and planners must decide where to install new batteries. Both decisions rest on how the many available demand forecasts—operators’ own forecasts, statistical models, machine learning, models driven by archived numerical weather forecasts, and pretrained foundation models—are combined. We propose *grouped decision-focused model averaging* (GDMA), which jointly learns a small number of latent segments of decision units and segment-specific combination weights by minimising the realised decision cost, with each unit’s own cost asymmetry, rather than forecast error; a soft version shrinks each unit’s weights towards its segment, and a fair version counts every unit’s relative improvement equally. We prove a finite-sample oracle inequality that separates the cost of learning segment memberships from the cost of learning weights, show that GDMA is asymptotically optimal and attains the risk of the infeasible unit-specific optimum when segments exist, and prove that it recovers the segments with probability approaching one. A planning stage turns the residual risk of the learned rule into an efficient or a fair siting of storage. On 125 grid areas in North America, Europe, Oceania, Asia and Africa with an average demand of 1.66 TW, using public hourly data from 2018 to 2026, soft GDMA lowers out-of-sample commitment cost by 18% worldwide (18–20% for the U.S. operators relative to their official forecasts), outperforms combining the same forecasts for accuracy and a decision-focused neural mixture-of-experts gate, and makes only 4% of grid areas worse off, with no systematic income gradient; its fair version is both cheaper and more equitable. Better combination substitutes for storage: without batteries, soft GDMA is cheaper than the status quo with 60 GW of optimally sited batteries, and it is worth 5–7 GW relative to standard combination rules. Siting storage on the status-quo error profile loses up to 15% of a small programme’s value, and maximin-fair siting raises the benefit of the worst-served continent at a price of fairness of about 1%.

Keywords: model averaging; decision-focused learning; latent groups; foundation models; fairness; capacity commitment; battery storage siting; energy management

1 Introduction

Electricity systems are managed through many local decision units that commit resources before uncertainty resolves. Every day, each of the balancing authorities, national system operators

and regional market operators that run the world’s grids must commit enough capacity to meet the next day’s hourly demand. Committing too little forces expensive real-time purchases, the activation of reserves or, in the worst case, load shedding; committing too much means paying for capacity that sits idle. The cost of a shortfall is not the same everywhere: an operator embedded in a dense web of interconnections can import from its neighbours, while an isolated system cannot. Winter Storm Uri in February 2021, when the weakly interconnected Texas grid shed load for days while better-connected neighbours coped, showed how much this asymmetry matters (Busby et al., 2021). At the same time, governments are committing to large battery-storage programmes, and planners must decide where those batteries should go.

Operators and planners do not lack forecasts. Operators publish their own day-ahead forecasts, simple persistence and calendar rules are always available, statistical and machine-learning models can be trained on the data, and pretrained foundation models now forecast any series zero-shot (Ansari et al., 2024). None dominates everywhere, which makes combining them attractive. How to combine them is less obvious. The forecast-combination and model-averaging literature, from Bates and Granger (1969) to the asymptotic theory of Hansen (2007), Hansen and Racine (2012), Zhang et al. (2013) and Zhang and Liu (2023), offers two options when there are many series. Weights can be estimated series by series, which respects heterogeneity but produces noisy weights when histories are short—the root of the “forecast-combination puzzle” that simple averages often beat estimated weights (Smith and Wallis, 2009; Claeskens et al., 2016). Or one set of weights can be estimated for all series, which is precise but imposes the same rule on systems that face different conditions. Deep learning offers a third option, a network that maps features to weights (Montero-Manso et al., 2020; Gong et al., 2026), at the price of thousands of parameters and weights that are hard to audit. All three typically judge a combination by its statistical accuracy, although the planner is judged by the cost of the resulting decisions; the decision-focused literature in operations management has shown that learning directly for the decision can be much better than forecasting accurately and then optimising (Ban and Rudin, 2019; Bertsimas and Kallus, 2020; Elmachtoub and Grigas, 2022).

This paper proposes *grouped decision-focused model averaging* (GDMA) and uses it to answer three management questions: how should grid operators on five continents combine the forecasts they have into day-ahead capacity commitments, who gains and who loses from a common planning rule, and where should new batteries be installed? GDMA assumes that decision units fall into a small number of latent segments whose members are best served by the same combination of candidate rules, and estimates the segments and the segment-specific weights jointly by minimising the realised system-wide decision cost, in money, with each unit’s own cost asymmetry. A soft version shrinks each unit’s own weights towards those of its segment. The number of segments and the shrinkage intensity are chosen by a time-ordered hold-out. Pooled weights, unit-specific weights and shrinkage towards pooled weights are special cases, and the combined commitment is always a convex combination of commitments the operator already understands. A second, planning stage takes the residual shortfall risk left by the learned rule and allocates a storage budget across grid areas.

We make five contributions.

1. **Method.** We formulate the joint learning of latent segments and combination weights

under an asymmetric, unit-specific decision loss, give a k -means-type algorithm with regret-space seeding in which every step is a small convex program, and link it to a storage-siting problem that is solved exactly by a greedy algorithm (Section 3). To our knowledge, neither the model-averaging, the decision-focused learning nor the latent-group panel literature (Bonhomme and Manresa, 2015; Su et al., 2016) has proposed this estimator.

2. **Learning theory.** We prove a finite-sample oracle inequality that separates the cost of learning memberships ($\log G/T$ per unit) from the cost of learning weights ($GM \log(NT)/(NT)$), which formalises why grouping beats unit-specific weights when histories are short; we show that GDMA is asymptotically optimal and, under a segment structure, attains the risk of the infeasible benchmark in which every unit knows its own optimal weights; that it recovers the segments with probability approaching one; and that on this event it coincides exactly with an estimator told the segments (Section 4).
3. **Evidence on five continents.** On 46 U.S. balancing authorities with their official forecasts (Section 6) and on a global panel of 125 grid areas in North America, Europe, Oceania, Asia and Africa with an average demand of 1.66 TW (Section 7), soft GDMA lowers the out-of-sample commitment cost by 18–20% relative to the best single forecaster, beats combining the same forecasts for accuracy, and beats a decision-focused neural mixture-of-experts gate by nearly ten percentage points worldwide; a pretrained foundation model and a model driven by archived day-ahead weather forecasts enter the candidate set on equal terms.
4. **Fairness.** We audit how the gains are distributed across grid areas and propose a fair variant that counts every area’s relative improvement equally. Soft GDMA makes 4% of areas worse off, against 13% for pooled weights and a third for the neural gate, shows no systematic gradient with national or state income, and its fair variant is both cheaper and more equitable (Section 7.4).
5. **Storage siting.** Better commitment rules substitute for storage: without any batteries, soft GDMA is cheaper than the status quo with 60 GW of optimally sited batteries, and it is worth 5–7 GW relative to standard combination rules. The optimal location of a storage programme depends on the commitment rule, siting on the status-quo error profile loses up to 15% of a small programme’s value, and maximin-fair siting helps the worst-served continent at a price of fairness of about 1% (Section 8).

The paper speaks to the theme of this special issue in both directions it describes: it develops a statistical learning method with guarantees, motivated by a concrete management problem, and it uses new, openly available data and models to answer a planning question of continental scale. Replication code and data-construction scripts are available from the author.

2 Related literature and the research gap

Model averaging and forecast combination. Since Bates and Granger (1969), combining forecasts has been one of the most reliable ways to improve predictive accuracy; Timmermann

(2006) and Wang et al. (2023) review five decades of work. Frequentist model averaging chooses weights by minimising an estimate of risk—a Mallows criterion (Hansen, 2007), leave-one-out or K -fold cross-validation (Hansen and Racine, 2012; Zhang and Liu, 2023), or forward validation for time series (Zhang and Zhang, 2023)—and establishes asymptotic optimality, i.e. that the chosen weights attain the lowest attainable risk. The framework now covers dependent data (Zhang et al., 2013), quantile losses (Lu and Su, 2015), general losses (Yu et al., 2024), time-varying weights (Sun et al., 2021), covariate-dependent weights (Gong et al., 2026), covariate shift (Zhang et al., 2026) and prediction intervals (Qu et al., 2025). The forecast-combination puzzle—that equal weights frequently beat estimated weights—is attributed mainly to estimation error in the weights (Smith and Wallis, 2009; Claeskens et al., 2016); remedies shrink estimated weights towards equal weights (Diebold and Shin, 2019) or restrict them (Zou et al., 2025). When many series must be forecast, weights are either estimated series by series or learned from series features (FFORMA, Montero-Manso et al., 2020); Aiolfi and Timmermann (2006) cluster *forecasting models* by past performance. All of these approaches either treat each target series separately or impose one rule on all of them, and none groups the *target units*.

Decision-focused learning. Operations research has long recognised that the loss that matters is the decision cost (Granger, 1969). The data-driven newsvendor literature learns order quantities directly from data (Levi et al., 2015; Ban and Rudin, 2019); prescriptive analytics maps covariates to decisions (Bertsimas and Kallus, 2020); and “smart predict-then-optimize” trains predictors on decision regret (Elmachtoub and Grigas, 2022); see Mandi et al. (2024) for a survey. In energy forecasting, Wang et al. (2019) combine probabilistic load forecasts by minimising pinball loss on the simplex, one series at a time. This literature trains single models or combines forecasts for one unit; it does not exploit cross-sectional structure when many heterogeneous units face related decisions.

Deep learning for forecasting and for combination weights. Pretrained time-series foundation models such as Chronos (Ansari et al., 2024) now forecast unseen series zero-shot with accuracy close to models trained on the target data, and deep networks are increasingly used to *produce* combination weights: FFORMA maps series features to weights with gradient boosting (Montero-Manso et al., 2020), and mixture-of-experts model averaging uses a softmax neural gate with covariate-dependent weights (Gong et al., 2026). These approaches are flexible, but they fit thousands of parameters, their weights are hard to audit, and—except for Gong et al. (2026), who study conditional generative models—they come without optimality guarantees for the combined decision. We include both a foundation model (as a candidate) and a decision-focused neural gate (as a competitor) in our application.

Storage siting. Where to place grid-scale batteries is usually decided with engineering models of the transmission network, solved by decomposition or stochastic programming (Piansky et al., 2024; Larroux et al., 2026). These models take the uncertainty that storage must absorb as an input. We make that input endogenous: the residual commitment risk depends on how operators combine their forecasts, so the value and the best location of storage depend on the forecasting policy.

Table 1: Positioning of GDMA relative to existing approaches.

Approach	Combines candidates	Decision-focused loss	Shares strength across units	Heterogeneous weights
Per-series combination / JMA / forward validation	✓			✓
Pooled combination	✓		✓	
Quantile / general-loss model averaging	✓	✓		✓
Shrinkage towards equal or pooled weights	✓		partial	✓
Feature-based weights (FFORMA)	✓		✓	✓
Neural mixture-of-experts gate	✓	(✓)	✓	✓
Data-driven newsvendor, SPO		✓	✓	
Latent-group panel models			✓	✓
GDMA (this paper)	✓	✓	✓	✓

Latent group structures. Panel econometrics models unobserved heterogeneity through latent groups estimated by k -means-type algorithms (Lin and Ng, 2012; Bonhomme and Manresa, 2015; Ando and Bai, 2016) or classifier-Lasso (Su et al., 2016). These methods group *regression parameters* under squared loss and focus on the parameters themselves, not on combining the outputs of existing forecasting systems under an asymmetric decision loss.

The gap. Table 1 summarises the position of GDMA. No existing method simultaneously (a) combines a library of existing candidate rules, including machine-learning and foundation-model forecasts, (b) chooses weights by the downstream decision cost with unit-specific cost asymmetries, and (c) shares statistical strength across many decision units through latent segments whose number is chosen from data, with guarantees on optimality and segment recovery; nor does any link the resulting operational risk to the siting of new storage. Filling this gap matters most when windows are short—because systems change and old data are stale—and units are many, which is exactly the situation of system-wide infrastructure planning.

3 Grouped decision-focused model averaging

3.1 The system-wide commitment problem

A planner is responsible for N decision units (in our application the balancing authorities of the U.S. grid). In period t unit i must commit a quantity q_{it} before demand y_{it} is realised. Committing too little costs $u_i > 0$ per unit of shortfall (emergency purchases, reserve activation or, in the limit, curtailed load); committing too much costs $o_i > 0$ per unit of excess (idle capacity that was paid for). The period cost is the newsvendor cost

$$C_i(q, y) = u_i (y - q)^+ + o_i (q - y)^+ = (u_i + o_i) \rho_{\tau_i}(y - q), \quad \tau_i = \frac{u_i}{u_i + o_i}, \quad (1)$$

where $\rho_\tau(r) = r(\tau - \mathbb{1}\{r < 0\})$ is the check function. The system cost that the planner cares about is $\sum_i \sum_t C_i(q_{it}, y_{it})$; under (1) the cost-minimising commitment is the conditional τ_i -quantile of demand (Gneiting, 2011). Critical ratios τ_i differ across units because the consequences of a shortfall differ: an operator that is tightly interconnected with its neighbours can import at moderate cost, while an isolated one cannot.

For every unit a library of M candidate commitment rules is available. Candidate m proposes $q_{it}^{(m)}$, computed from information available before period t ; we write $\mathbf{q}_{it} = (q_{it}^{(1)}, \dots, q_{it}^{(M)})^\top$. The candidates may come from any source: an operator’s own published forecast, simple persistence rules, statistical models or machine-learning models, each turned into a commitment by adding its own τ_i -quantile residual margin. The planner combines them linearly,

$$q_{it}(\mathbf{w}) = \mathbf{q}_{it}^\top \mathbf{w}, \quad \mathbf{w} \in \Delta_M = \left\{ \mathbf{w} \in \mathbb{R}^M : w_m \geq 0, \sum_m w_m = 1 \right\}.$$

The simplex constraint keeps the combined commitment inside the range spanned by the candidates, which is what makes the combination interpretable and auditable by operators, and it is known to stabilise estimated weights (Zou et al., 2025).

3.2 Three ways to estimate the weights

Let the estimation window contain periods $t = 1, \dots, T$ for which both candidate commitments and realised demand are observed (a forward-validation sample in the terminology of Zhang and Zhang, 2023). Write $\hat{R}_i(\mathbf{w}) = T^{-1} \sum_{t=1}^T C_i(\mathbf{q}_{it}^\top \mathbf{w}, y_{it})$ for the empirical cost of unit i .

Unit-specific weights. $\hat{\mathbf{w}}_i = \arg \min_{\mathbf{w} \in \Delta_M} \hat{R}_i(\mathbf{w})$. This respects heterogeneity but estimates $M - 1$ free parameters from T serially dependent observations per unit; it is the decision-focused counterpart of per-series forecast combination, and suffers from the same estimation noise that underlies the forecast-combination puzzle (Smith and Wallis, 2009; Claeskens et al., 2016).

Pooled weights. $\hat{\mathbf{w}} = \arg \min_{\mathbf{w}} \sum_i \hat{R}_i(\mathbf{w})$. This uses all NT observations but forces a single policy on units that may need different ones.

Grouped weights (proposed). We posit that units fall into a small number G of latent segments whose members are well served by the same combination. The GDMA estimator solves

$$(\hat{\gamma}, \hat{\mathbf{W}}) = \arg \min_{\gamma \in \{1, \dots, G\}^N, \mathbf{W} = (\mathbf{w}_1, \dots, \mathbf{w}_G) \in \Delta_M^G} \sum_{i=1}^N \sum_{t=1}^T C_i(\mathbf{q}_{it}^\top \mathbf{w}_{\gamma_i}, y_{it}), \quad (2)$$

and unit i commits $\mathbf{q}_{i,T+1}^\top \hat{\mathbf{w}}_{\hat{\gamma}_i}$. $G = 1$ gives pooled weights and $G = N$ gives unit-specific weights, so the number of segments indexes the whole range between the two extremes. Three features separate (2) from existing proposals. First, the criterion is the decision cost in the planner’s own units (money or energy), so units are grouped by *how they should act*, not by how their forecasts look. Second, memberships and weights are estimated jointly; a two-step procedure that clusters noisy unit-specific weights and then re-estimates is a natural but inferior alternative (Section 5). Third, because each unit keeps its own critical ratio and its own candidate margins, units in one segment share a combination rule but not a commitment level.

3.3 Computation

Problem (2) is a mixed-integer problem, but it has the structure of k -means with a non-Euclidean, data-dependent distortion, as in the grouped fixed-effects estimator of Bonhomme

Algorithm 1 GDMA for a fixed number of segments G

Require: candidate commitments $\{q_{it}\}$, demand $\{y_{it}\}$, costs $\{u_i, o_i\}$, G , number of starts S

- 1: compute unit-specific weights $\hat{\mathbf{w}}_i$ and the regret matrix $D_{ij} = \hat{R}_i(\hat{\mathbf{w}}_j) - \hat{R}_i(\hat{\mathbf{w}}_i)$
 - 2: **for** $s = 1, \dots, S$ **do**
 - 3: draw seeds $j_1, \dots, j_G$ by k -means++ on D and set $\mathbf{w}_g \leftarrow \hat{\mathbf{w}}_{j_g}$
 - 4: **repeat**
 - 5: $\gamma_i \leftarrow \arg \min_g \hat{R}_i(\mathbf{w}_g)$ for all i ▷ assignment step
 - 6: re-seed any empty segment with the unit of largest regret
 - 7: $\mathbf{w}_g \leftarrow \arg \min_{\mathbf{w} \in \Delta_M} \sum_{i: \gamma_i = g} \hat{R}_i(\mathbf{w})$ for all g ▷ pooled convex step
 - 8: **until** memberships do not change
 - 9: **end for**
 - 10: **return** the solution with the lowest objective (2)
-

and Manresa (2015). Given memberships, the weights of each segment solve a pooled convex problem on the simplex; given weights, each unit joins the segment whose weights give it the lowest in-window cost. Algorithm 1 alternates the two steps. The weight step is a linear program; for speed we minimise a softplus-smoothed check function, $\rho_\tau^\varepsilon(r) = \tau r + \varepsilon \log(1 + e^{-r/\varepsilon})$, with bandwidth ε equal to 2% of the unit’s typical decision error, by sequential quadratic programming. On our data the smoothed solution is within 0.01% of the exact linear-programming optimum in cost. Each iteration weakly decreases the objective, so the algorithm converges to a local solution in finitely many steps; we run several starts and keep the best. Starting values are drawn by a k -means++ rule (Arthur and Vassilvitskii, 2007) in *regret space*: the next seed unit is drawn with probability proportional to the extra cost that units would incur if forced to use the closest existing seed’s own weights. In rolling applications the memberships of the previous window provide an additional warm start.

3.4 Soft segments

Segments need not be exact. When units within a segment are similar but not identical, a unit may benefit from a rule that sits between its own estimated weights and those of its segment. *Soft GDMA* uses

$$\hat{\mathbf{w}}_i^{\text{soft}} = (1 - \kappa) \hat{\mathbf{w}}_i + \kappa \hat{\mathbf{w}}_{\gamma_i}, \quad \kappa \in [0, 1], \quad (3)$$

where $\hat{\mathbf{w}}_i$ are the unit-specific weights and $\hat{\mathbf{w}}_{\gamma_i}$ the GDMA weights of the unit’s segment. Because the simplex is convex, $\hat{\mathbf{w}}_i^{\text{soft}} \in \Delta_M$. The family nests every benchmark: $\kappa = 1$ is GDMA, $\kappa = 0$ is unit-specific weights, $G = 1$ is shrinkage towards pooled weights (a decision-focused analogue of Diebold and Shin, 2019), and $G = 1, \kappa = 1$ is pooled weights. The number of segments is chosen exactly as for GDMA (next subsection); given G , κ is chosen on the same hold-out block from $\{0, 0.25, 0.5, 0.75, 1\}$. Choosing G first keeps the search small, which matters because hold-out costs are noisy when windows are short. Segment centres play the role of data-driven prior means, in the spirit of empirical Bayes shrinkage towards group-specific targets.

3.5 Choosing the number of segments

We split the estimation window into an early block (the first two thirds) and a late block. For each $G \in \{1, \dots, G_{\max}\}$ we estimate (2) on the early block and record every unit’s cost on the

late block under its estimated membership and weights. Let $G^\dagger$ minimise the total late-block cost. Following the one-standard-error rule (Hastie et al., 2009), we choose the smallest G whose total late-block cost exceeds that of $G^\dagger$ by no more than one standard error of the summed unit-level cost differences, and then re-estimate on the full window. The rule respects time order and prefers parsimonious segmentations, which are easier to explain to the operators who would have to adopt them.

3.6 From commitment rules to storage siting

The combination rule is an operational decision, re-estimated every week. A planner also faces a slower, capital-intensive decision: where to install new flexible capacity. We link the two with a two-stage design. In the operational stage, the learned rule leaves each unit with hourly shortfalls $s_{it} = (y_{it} - q_{it})^+$ and surpluses $x_{it} = (q_{it} - y_{it})^+$. In the planning stage, a battery with power P_i (MW) and energy $\bar{h}P_i$ (MWh) installed in unit i discharges to cover shortfalls and recharges from committed-but-unused capacity, which has already been paid for. With state of charge e_{it} and round-trip efficiency η , the chronological dispatch is

$$d_{it} = \min\{P_i, s_{it}, e_{it}\}, \quad c_{it} = \min\{P_i, x_{it}, (\bar{h}P_i - e_{it} + d_{it})/\eta\},$$

$$e_{i,t+1} = e_{it} - d_{it} + \eta c_{it},$$

and every MWh discharged avoids the underage cost u_i . The annual value of battery power, $V_i(P)$, equal to u_i times the expected energy discharged per year, is non-decreasing and typically concave in P , because each additional MW of power is called on only in rarer and larger shortfalls; where the estimated curve is not concave on the grid we replace later increments by the running minimum of the marginal values, which yields a concave lower bound. Given an annualised cost κ per MW of battery power, the planner solves

$$\max_{P_1, \dots, P_N \geq 0} \sum_{i=1}^N \{V_i(P_i) - \kappa P_i\} \quad \text{subject to} \quad \sum_{i=1}^N P_i \leq B, \quad (4)$$

where B is a storage budget or deployment target (for example a national or continental procurement programme). Because each (concavified) V_i is concave, a greedy algorithm that adds power in small increments to the unit with the highest marginal value is optimal for (4); without the budget constraint, each unit simply installs power until its marginal value falls to κ . We estimate V_i on a grid of battery sizes from the out-of-sample shortfall and surplus paths of each commitment rule.

The two stages interact in two ways that matter for management. First, better commitment rules shrink the shortfalls that storage must cover, so part of the storage requirement can be met by better forecast combination instead of new hardware; we measure this *storage equivalent* of better forecasting as the battery power that the status-quo rule needs to reach the cost of the better rule without storage. Second, the siting decision depends on the rule: batteries placed where the status-quo rule is weakest may be placed where they add little once operations improve.

3.7 Fairness across decision units

At continental scale the objective in (2) has a distributional consequence. Because costs are measured in money or energy, the largest systems dominate the criterion, and a segment rule that is excellent for a large system may be mediocre for a small or data-poor one that happens to share its segment. The same holds for storage: problem (4) places batteries where the value is highest in absolute terms. Energy-justice scholarship stresses that the distribution of the benefits of energy infrastructure matters in its own right (Sovacool and Dworkin, 2015), and the operations literature measures the efficiency cost of fair allocations by the *price of fairness* (Bertsimas et al., 2011). We therefore add two fair variants and audit every method for how its benefits are distributed.

Fair soft GDMA (one area, one vote). Let c_i^0 be the in-window cost of unit i under its single best forecaster. Fair soft GDMA solves the soft GDMA problem with every unit’s costs divided by c_i^0 (normalised to mean one), that is, with u_i and o_i replaced by $\omega_i u_i$ and $\omega_i o_i$, $\omega_i \propto 1/c_i^0$. Each unit then counts in proportion to its *relative* improvement, so that a 10% saving in a 70 MW municipal utility weighs as much as a 10% saving in a 90 GW regional market. This is the relative-cost analogue of group-robust learning (Sagawa et al., 2020); it changes only the weights of the criterion, not the algorithm.

Fair storage siting. We consider two fair versions of (4) over continents c (any grouping, such as countries or income groups, works the same way). *Proportional floors* require every continent to receive at least a share λ of its share of demand in the budget; the floors are filled greedily within each continent and the rest of the budget greedily across all units. *Maximin siting* allocates each increment of battery power to the continent whose relative benefit—the value captured divided by its commitment cost without storage—is currently the lowest, and within it to the unit with the highest marginal value, which implements a max–min notion of fairness (Bertsimas et al., 2011). The price of fairness is the share of the efficient programme’s value that each fair version gives up.

Audit. For every method we report the distribution across units of the ratio of its cost to the unit’s own single best forecaster, the share of units made worse off, the median ratio of the worst-served continent, and separate ratios for data-rich and data-poor systems.

4 Theoretical properties

This section studies the global minimiser of (2) for a given G . We show that (i) its expected decision cost is close to that of the best grouped policy, with an estimation error that separates the price of learning memberships from the price of learning weights; (ii) it is asymptotically optimal and, under a latent segment structure, attains the risk of the infeasible unit-specific oracle; and (iii) it recovers the segments with probability approaching one, in which case its weights coincide with those of an estimator that is told the segments. Proofs are in Appendix A.

4.1 Setting and assumptions

Let $\mathbf{z}_{it} = (y_{it}, \mathbf{q}_{it})$. Define the population risk of unit i , $R_i(\mathbf{w}) = \mathbb{E} C_i(\mathbf{q}_{it}^\top \mathbf{w}, y_{it})$, and for a grouped policy $(\gamma, \mathbf{W})$ the average risk and its empirical analogue

$$R(\gamma, \mathbf{W}) = \frac{1}{N} \sum_{i=1}^N R_i(\mathbf{w}_{\gamma_i}), \quad \hat{R}(\gamma, \mathbf{W}) = \frac{1}{NT} \sum_{i=1}^N \sum_{t=1}^T C_i(\mathbf{q}_{it}^\top \mathbf{w}_{\gamma_i}, y_{it}).$$

Let $\mathcal{P}_G = \{1, \dots, G\}^N \times \Delta_M^G$ and $R_G^* = \inf_{\mathcal{P}_G} R$.

Assumption 1 (Stationarity and boundedness). *For each i , $\{\mathbf{z}_{it}\}_t$ is strictly stationary, $|y_{it}| \leq B$ and $\|\mathbf{q}_{it}\|_\infty \leq B$ almost surely, and $\max_i(u_i + o_i) \leq \bar{a}$.*

Stationarity refers to the candidate commitments as well as demand; it holds, for instance, when every candidate is computed from a rolling window of fixed length of a stationary process. Boundedness is natural once demand is normalised by a unit's scale. Under Assumption 1 each loss lies in $[0, L]$ with $L = 2\bar{a}B$ and is Lipschitz in the weights: $|C_i(\mathbf{q}^\top \mathbf{w}, y) - C_i(\mathbf{q}^\top \mathbf{w}', y)| \leq \bar{a}B \|\mathbf{w} - \mathbf{w}'\|_1$.

Assumption 2 (Concentration). *There are constants $c > 0$ and an effective sample length $T_e \leq T$ such that, for every fixed $(\gamma, \mathbf{W}) \in \mathcal{P}_G$ and every $x > 0$,*

$$\Pr(|\hat{R}(\gamma, \mathbf{W}) - R(\gamma, \mathbf{W})| > x) \leq 2 \exp(-c N T_e x^2 / L^2),$$

and, for every i and fixed $\mathbf{w}$, $\Pr(|\hat{R}_i(\mathbf{w}) - R_i(\mathbf{w})| > x) \leq 2 \exp(-c T_e x^2 / L^2)$.

Assumption 2 is a high-level condition. It holds with $T_e = T$ when observations are independent (Hoeffding's inequality) and, when units are independent and each $\{\mathbf{z}_{it}\}_t$ is geometrically β -mixing, with $T_e = T/(C \log T)$ by the blocking argument of Yu (1994) or the Bernstein inequality of Merlevède et al. (2011). Cross-sectional dependence through common shocks (weather, in our application) reduces the effective number of independent units; the results continue to hold with N replaced by an effective cross-sectional size.

4.2 An oracle inequality

Theorem 1 (Oracle inequality). *Let Assumptions 1–2 hold and let $(\hat{\gamma}, \hat{\mathbf{W}})$ be a global minimiser of (2). For any $\delta \in (0, 1)$, with probability at least $1 - \delta$,*

$$R(\hat{\gamma}, \hat{\mathbf{W}}) \leq R_G^* + 2L \sqrt{\frac{GM \log(3NT_e) + N \log G + \log(2/\delta)}{c N T_e}} + \frac{4\bar{a}B}{N T_e}. \quad (5)$$

The square-root term has a transparent decomposition. Per unit, learning the segment membership costs $\log G/T_e$, while learning the segment weights costs $GM \log(NT_e)/(NT_e)$. The same argument applied to unit-specific weights ($G = N$) gives $M \log(NT_e)/T_e$, and applied to pooled weights ($G = 1$) gives $M \log(NT_e)/(NT_e)$ but a larger approximation error $R_1^* - R_N^*$. Grouping therefore replaces the M per-unit parameters of unit-specific weights by a single discrete label that costs $\log G$ rather than $M \log(NT_e)$ to learn. This is the formal sense in which GDMA resolves the tension between the noise of unit-specific weights and the bias of

pooled weights: with $M = 7$ candidates, $N = 46$ units, $G = 4$ segments and an effective length of $T_e = 28$ (one observation per day of a four-week window), the variance proxy under the square root is 0.23 for GDMA against 2.07 for unit-specific weights, so the bound is about three times tighter, while the approximation error of GDMA is zero whenever the segment structure is correct.

4.3 Asymptotic optimality

Following the model-averaging literature (Hansen, 2007; Hansen and Racine, 2012; Zhang and Liu, 2023), call a weight choice asymptotically optimal if the ratio of its risk to the lowest attainable risk in its class tends to one.

Corollary 1 (Asymptotic optimality). *Let the assumptions of Theorem 1 hold, with G and M possibly growing, and suppose $R_G^* \geq \kappa > 0$. If $GM \log(NT_e)/(NT_e) \rightarrow 0$ and $\log G/T_e \rightarrow 0$, then $R(\hat{\gamma}, \hat{\mathbf{W}})/R_G^* \rightarrow_p 1$. If, in addition, there are memberships γ^0 with $G_0 \leq G$ segments such that $\arg \min_{\mathbf{w}} R_i(\mathbf{w})$ depends on i only through γ_i^0 , then $R_G^* = R_N^* = N^{-1} \sum_i \min_{\mathbf{w}} R_i(\mathbf{w})$ and*

$$\frac{R(\hat{\gamma}, \hat{\mathbf{W}})}{N^{-1} \sum_i \min_{\mathbf{w} \in \Delta_M} R_i(\mathbf{w})} \rightarrow_p 1.$$

The second statement is the relevant one for practice: under a segment structure, GDMA attains the risk of the infeasible benchmark in which every unit knows its own optimal weights, although it estimates only $G(M - 1)$ continuous parameters. Unit-specific weights attain the same limit only if $M \log(NT_e)/T_e \rightarrow 0$, a much stronger requirement when windows are short because the system is non-stationary and old data are of limited use. The condition $R_G^* \geq \kappa$ excludes the degenerate case of perfect foresight, which never occurs in commitment problems.

4.4 Recovery of the segments

Assumption 3 (Segment structure). (a) *There are $\gamma^0 \in \{1, \dots, G\}^N$ and distinct $\mathbf{w}_1^0, \dots, \mathbf{w}_G^0 \in \Delta_M$ such that, for all i and all $\mathbf{w} \in \Delta_M$, $R_i(\mathbf{w}) - R_i(\mathbf{w}_{\gamma_i^0}^0) \geq \lambda \|\mathbf{w} - \mathbf{w}_{\gamma_i^0}^0\|_2^2$ for some $\lambda > 0$.* (b) *$\min_{g \neq h} \|\mathbf{w}_g^0 - \mathbf{w}_h^0\|_2 \geq d > 0$ and every segment contains at least πN units, $\pi > 0$.*

Part (a) is a quadratic-growth condition. It holds when the conditional density of demand given the candidates is bounded away from zero near the optimal commitment and the candidates are not collinear, because the Hessian of R_i is then $(u_i + o_i) \mathbb{E}[f_{y|q}(\mathbf{q}^\top \mathbf{w}) \mathbf{q} \mathbf{q}^\top]$, which is positive definite on the tangent space of the simplex. Part (b) requires segments to be genuinely different and non-negligible, as in Bonhomme and Manresa (2015).

Theorem 2 (Segment recovery and oracle equivalence). *Let Assumptions 1–3 hold with G equal to the true number of segments, and let $N, T_e \rightarrow \infty$ with $\log N/T_e \rightarrow 0$. Then, up to a relabelling of segments, (i) $\max_g \|\hat{\mathbf{w}}_g - \mathbf{w}_g^0\|_2 \rightarrow_p 0$; (ii) $\Pr(\hat{\gamma} \neq \gamma^0) \leq N C_M \exp(-c' T_e) + o(1) \rightarrow 0$ for constants $C_M, c' > 0$ that do not depend on N or T ; (iii) with probability approaching one, $\hat{\mathbf{W}}$ equals the infeasible estimator $\tilde{\mathbf{W}} = \arg \min_{\mathbf{W}} \hat{R}(\gamma^0, \mathbf{W})$ that knows the segments.*

Part (iii) is an exact equivalence, not an asymptotic approximation: on the event that memberships are recovered, the joint minimiser and the known-segment minimiser are the same

object. Any distributional result available for the known-segment estimator therefore transfers to GDMA without modification. The recovery condition $\log N/T_e \rightarrow 0$ is mild: the number of units may grow polynomially in the window length.

Soft GDMA. For fixed (G, κ) , R_i is convex in $\mathbf{w}$, so the risk of (3) satisfies $R_i(\hat{\mathbf{w}}_i^{\text{soft}}) \leq (1 - \kappa)R_i(\hat{\mathbf{w}}_i) + \kappa R_i(\hat{\mathbf{w}}_{\hat{\gamma}_i})$. Averaging over units, the excess risk of soft GDMA is at most the κ -weighted average of the excess risks of unit-specific weights and of GDMA, each controlled by Theorem 1. Under a segment structure, therefore, any κ bounded away from zero inherits the faster rate of GDMA for the κ share of the combination, while $\kappa < 1$ buys robustness when segments are only approximate. Choosing (G, κ) from a finite grid by hold-out adds only a $\log(|\mathcal{G}|)$ term to the hold-out error, where $|\mathcal{G}|$ is the grid size.

Weighted and fair criteria. All results of this section hold for criteria in which unit i 's costs are multiplied by a weight $\omega_i \in [\underline{\omega}, \bar{\omega}]$ with $0 < \underline{\omega} \leq \bar{\omega} < \infty$: the weights only rescale u_i and o_i , so Assumptions 1–3 hold with $\bar{a}$ replaced by $\bar{\omega}\bar{a}$ and λ by $\underline{\omega}\lambda$. Fair soft GDMA uses data-dependent weights $\omega_i \propto 1/c_i^0$; treating them as fixed within a window, Theorem 1 bounds its weighted excess risk. Under the segment structure of Corollary 1, the weighted and unweighted criteria have the same minimiser, because every unit attains its own optimal weights, and since each unit's excess risk is non-negative the unweighted excess risk is at most $1/\underline{\omega}$ times the weighted one (weights normalised to mean one). Fairness then costs nothing asymptotically. When segments are only approximate, the fair criterion trades some total cost for a more even distribution of the gains; this is the price of fairness that we measure empirically.

5 Monte Carlo evidence

5.1 Design

We simulate $N \in \{50, 200\}$ units and $M = 5$ candidate commitment rules. Unit i has demand $y_{it} = \mu_i + x_{it}$, with $\mu_i \sim U[20, 100]$ and x_{it} a stationary AR(1) process with coefficient 0.5 and standard deviation 5, and critical ratio $\tau_i \sim U[0.6, 0.95]$ ($u_i = \tau_i/(1 - \tau_i)$, $o_i = 1$). Candidate m forecasts $f_{itm} = y_{it} - e_{itm}$, where the errors follow AR(1) processes with coefficient 0.5 and innovations with unit-specific scales s_{im} and skewness signs $k_{im} \in \{-1, 0, +1\}$ (a centred gamma distribution for ± 1 , a normal distribution for 0). Each candidate becomes a commitment by adding the τ_i -quantile of its errors in a pre-sample of 200 periods. Three designs define the heterogeneity: *grouped*—every unit belongs to one of three segments with segment-specific error profiles (s_i, k_i) , so that different candidates are best in different segments; *homogeneous*—all units share the first profile; *continuous*—each unit's error scales are a Dirichlet(1,1,1) mixture of the three profiles, so that no two units share their optimal weights and GDMA's segment structure is misspecified. Weights are estimated on $T \in \{48, 168, 672\}$ periods and evaluated on 1,000 new periods. We report the out-of-sample cost relative to an infeasible oracle that uses, for every unit, the exact cost-minimising weights on the evaluation sample itself. Each cell uses 50 replications.

Table 2: Monte Carlo results: out-of-sample newsvendor cost relative to the unit-specific oracle (means over 50 replications), average selected number of segments $\bar{G}$ of GDMA, and adjusted Rand index (ARI; Hubert and Arabie, 1985) between estimated and true segments for GDMA and two-step clustering. Bold: lowest in the row.

Design	N	T	Out-of-sample cost / oracle							$\bar{G}$	ARI	
			EW	Pooled	Per-unit	Shrink	2-step	GDMA	Soft		GDMA	2-step
Grouped	50	48	1.496	1.414	1.084	1.090	1.055	1.050	1.041	3.4	0.87	0.75
	50	168	1.491	1.410	1.033	1.033	1.031	1.030	1.023	3.7	0.88	0.79
	50	672	1.481	1.398	1.013	1.013	1.019	1.018	1.011	5.3	0.68	0.67
	200	48	1.491	1.418	1.086	1.089	1.049	1.040	1.035	3.1	0.93	0.79
	200	168	1.492	1.419	1.032	1.032	1.028	1.023	1.019	5.2	0.71	0.73
	200	672	1.491	1.416	1.012	1.012	1.021	1.017	1.011	5.8	0.62	0.66
Homogeneous	50	48	1.321	1.027	1.097	1.027	1.032	1.028	1.028	1.1	—	—
	50	168	1.322	1.025	1.036	1.020	1.026	1.024	1.021	1.3	—	—
	50	672	1.322	1.025	1.015	1.013	1.017	1.016	1.012	2.6	—	—
	200	48	1.321	1.026	1.094	1.025	1.027	1.026	1.025	1.0	—	—
	200	168	1.320	1.025	1.038	1.020	1.026	1.024	1.020	1.3	—	—
	200	672	1.321	1.025	1.014	1.013	1.017	1.015	1.012	2.9	—	—
Continuous	50	48	1.206	1.148	1.091	1.075	1.090	1.091	1.068	2.7	—	—
	50	168	1.206	1.148	1.035	1.036	1.051	1.055	1.031	4.0	—	—
	50	672	1.203	1.145	1.014	1.014	1.035	1.035	1.014	5.4	—	—
	200	48	1.202	1.146	1.094	1.075	1.090	1.082	1.062	3.1	—	—
	200	168	1.205	1.149	1.035	1.038	1.050	1.049	1.030	4.9	—	—
	200	672	1.206	1.150	1.013	1.013	1.040	1.037	1.014	5.9	—	—

5.2 Results

Table 2 supports the theory and clarifies when grouping pays.

Soft GDMA is the most reliable method. It has the lowest cost, or is within 0.1 percentage points of the lowest, in every one of the 18 cells. Its advantage is largest when windows are short: with true segments and $T = 48$ it is 3.5–4.1% above the oracle against 8.4–8.6% for unit-specific weights, and even under continuous heterogeneity, where segments are misspecified, it beats unit-specific weights (6.2–6.8% against 9.1–9.4% above the oracle) and shrinkage towards pooled weights (7.5%). When units share one error profile it matches pooled weights for short windows and beats them for long ones, where the heterogeneity in critical ratios makes some unit-level adjustment worthwhile.

Hard segments trade bias for variance exactly as Theorem 1 predicts. Plain GDMA beats unit-specific weights in 96–100% of the replications with true segments and $T = 48$, and in all replications with homogeneous units for $T \leq 168$. With long windows ($T = 672$), unit-specific weights are estimated precisely and become slightly better (1.2–1.3% against 1.7–1.8% above the oracle). Here GDMA selects five to six segments rather than three, because units within a true segment have different critical ratios and therefore different optimal weights: with enough data the hold-out rule correctly detects this finer structure, which is also why the ARI falls from about 0.9 at $T = 48$ to about 0.65 at $T = 672$. Soft GDMA removes this residual bias through unit-level shrinkage and is never worse than unit-specific weights by more than 0.1 percentage points.

Joint estimation recovers segments better than clustering estimated weights. With short windows, the ARI of GDMA is 0.87–0.93, against 0.75–0.79 for two-step k -means on noisy unit-

level weights, and GDMA’s cost is lower in every grouped cell with $T \leq 168$. When units share one error profile, the one-standard-error rule selects a single segment in most replications with $T \leq 168$ ($\bar{G} = 1.0\text{--}1.3$).

6 Day-ahead capacity commitment on the U.S. grid

6.1 Data

We use the complete public record of the U.S. Energy Information Administration’s Hourly Electric Grid Monitor (Form EIA-930; [U.S. Energy Information Administration, 2026](#)), downloaded on 29 September 2026. For every balancing authority (BA) in the lower 48 states the file reports hourly demand, the BA’s own day-ahead demand forecast and total interchange with neighbouring BAs. We use the period 1 January 2019 to 26 September 2026 in UTC hours and exclude the fourteen regional aggregates. Following [Ruggles et al. \(2020\)](#), non-positive values and values outside $[0.3, 3]$ times a centred seven-day rolling median are treated as missing, and gaps of up to three hours are interpolated. We keep BAs with at least 97% valid demand, at least 93% valid official forecasts, mean demand above 50 MW and a median absolute official forecast error below 15% of median demand. Forty-six BAs remain (Table 16 in Appendix C); they range from PJM (mean demand 92.5 GW) to the City of Homestead (71 MW), include all seven independent system operators, and account for 97% of lower-48 demand. Ten BAs are dropped, mostly small systems with long gaps or implausible official forecasts; 0.27% of the retained BA-hours are missing.

6.2 Decision problem and cost calibration

Before each operating day d , BA i commits hourly capacity $q_{i,d,h}$, $h = 0, \dots, 23$. The information set contains realised demand up to the end of day $d-1$ and the BA’s published day-ahead forecast for day d . Excess commitment costs $o_i = 1$ per MWh, which normalises all costs to units of the per-MWh cost of idle committed capacity. Shortfalls cost $u_i = \tau_i/(1 - \tau_i)$, where the critical ratio τ_i reflects the BA’s ability to import. We measure interchange flexibility in the pre-sample year 2019 as the range between the 5th and 95th percentiles of hourly total interchange, divided by mean demand, and let τ_i decline linearly in the flexibility rank from 0.95 for the least flexible BA to 0.80 for the most flexible one. ERCOT, which is almost islanded, receives $\tau = 0.95$ ($u_i = 19$); the Chelan County PUD in the densely interconnected Pacific Northwest receives $\tau = 0.80$ ($u_i = 4$). This calibration is deliberately simple and transparent; Section 6.7 shows that the ranking of methods is the same under common critical ratios between 0.5 and 0.95.

6.3 Candidate rules

Seven candidate forecasts are produced for every BA, day and hour, each using only information available before the operating day: (1) the BA’s official day-ahead forecast; (2) persistence (same hour of $d-1$); (3) weekly persistence (same hour of $d-7$); (4) the mean of the same hour over $d-1, \dots, d-7$; (5) the mean of the same hour on $d-7$, $d-14$ and $d-21$; (6) a BA-and-hour-specific linear regression on the $d-1$ and $d-7$ values, the official forecast and weekday

dummies, estimated on a rolling 56-day window and refitted weekly; and (7) a single global LightGBM model (Ke et al., 2017) across all BAs and hours, trained on lags, calendar variables and the official forecast, all scaled by the BA’s recent mean demand, refitted every 28 days on the preceding 365 days. Model (7) is the kind of machine-learning forecaster a central planner could run for every BA at low cost. Each forecast $f^{(m)}$ becomes a candidate commitment by adding its own safety margin,

$$q_{i,d,h}^{(m)} = f_{i,d,h}^{(m)} \{1 + \hat{Q}_{\tau_i}(e_i^{(m)})\},$$

where $\hat{Q}_{\tau_i}(e_i^{(m)})$ is the empirical τ_i -quantile of the relative errors $(y - f^{(m)}) / \max\{f^{(m)}, 0.2\bar{y}_i\}$ of forecast m for BA i over the previous 28 days; the floor at 20% of the BA’s mean demand $\bar{y}_i$ prevents an occasional implausibly small forecast from producing an unbounded margin. The candidate rules are available from February 2020, so the evaluation period runs from 1 April 2020 to 26 September 2026: 2,342 days, 335 weekly re-estimations and 2.58 billion MWh of demand.

6.4 Competing methods

Every seven days each method re-estimates its combination from the most recent W days, $W \in \{7, 14, 28, 56\}$ (each day contributes 24 hourly observations per BA) and commits for the following seven days. Besides GDMA and soft GDMA ($G_{\max} = 6$) we consider: the official-forecast policy (candidate 1 alone), which is the status quo; equal weights over the seven candidates; the best single candidate per BA over the window; pooled decision-focused weights; per-BA decision-focused weights (the decision-focused analogue of per-series forward validation); shrinkage of per-BA weights towards pooled weights with the shrinkage intensity chosen on the hold-out (a decision-focused version of Diebold and Shin, 2019); two-step clustering, which runs k -means on the per-BA weight vectors and re-estimates pooled weights within clusters, with the number of clusters chosen by the same hold-out; and two *forecast-then-commit* methods, which estimate per-BA or grouped combination weights by minimising scaled squared forecast error and then add the τ_i -quantile of the combined forecast’s relative errors. The last two isolate the value of estimating the combination for the decision rather than for accuracy.

6.5 Results

Table 3 contains the main result. Four findings stand out.

First, deciding for the decision pays. Soft GDMA lowers the system-wide commitment cost by 18–20% relative to the status quo of committing on the official forecasts, and every decision-focused combination method lowers it by at least 15%; all reductions are significant at the 1% level. The forecast-then-commit methods, which combine the same seven forecasts for accuracy and then add a quantile margin, save only 3–18%, and they are significantly more expensive than soft GDMA for every window length. The gap is largest when windows are short ($W = 7$: 0.971 and 0.907 against 0.817), because accuracy-optimal weights are chosen without regard to the asymmetric costs and the margin estimated from one week of combined errors is noisy. Equal weights are the worst choice: they give weight to the naive calendar rules and raise cost

Table 3: Out-of-sample system-wide commitment cost, 1 April 2020 to 26 September 2026, relative to committing on the official day-ahead forecasts. W is the length of the estimation window. Bold: lowest cost in the column. Stars: Diebold–Mariano test of equal daily system cost against soft GDMA with Newey–West variance (7 lags); *, **, ***: $p < 0.10, 0.05, 0.01$.

Method	System cost relative to the official-forecast policy				Share of BAs improved ^a
	$W = 7$ days	$W = 14$ days	$W = 28$ days	$W = 56$ days	
Official forecast	1.0000***	1.0000***	1.0000***	1.0000***	0.00
Equal weights	1.5287***	1.5287***	1.5287***	1.5287***	0.13
Best single (per BA)	0.8490***	0.8288***	0.8245***	0.8234***	0.76
Pooled DF weights	0.8155	0.8110***	0.8075***	0.8039***	0.78
Per-BA DF weights	0.8496***	0.8221***	0.8069***	0.8007***	0.78
Shrinkage to pooled	0.8154	0.8053	0.8009	0.7962	0.80
Two-step k -means	0.8242	0.8077*	0.8041***	0.7980	0.83
Forecast-then-commit, per BA	0.9713***	0.8850***	0.8382***	0.8233***	0.65
Forecast-then-commit, grouped	0.9068***	0.8523***	0.8260***	0.8277***	0.67
GDMA	0.8162	0.8077***	0.8042***	0.7999***	0.78
GDMA, minimum hold-out rule	0.8227***	0.8108***	0.8032***	0.7977*	0.80
Soft GDMA	0.8163	0.8048	0.7999	0.7965	0.80

^a Share of the 46 BAs whose own cost is lower than under the official-forecast policy ($W = 28$).

by 53%.

Second, neither extreme is right. Per-BA weights are expensive when windows are short (0.850 at $W = 7$, 4.2% above pooled weights), pooled weights are expensive when windows are long (0.804 at $W = 56$, 0.9% above soft GDMA), and the per-window ranking of the two reverses between $W = 14$ and $W = 28$. This is the bias–variance trade-off of Theorem 1 in action.

Third, soft GDMA is the most reliable method. It has the lowest system cost for $W = 14$ and $W = 28$ and is within 0.1% of the best method for $W = 7$ and $W = 56$; no method is significantly cheaper than soft GDMA for any window, while it is significantly cheaper than pooled weights, per-BA weights, the forecast-then-commit methods and plain GDMA for most windows. Its closest competitor is decision-focused shrinkage towards pooled weights, which is the special case $G = 1$ of soft GDMA and is statistically indistinguishable from it; the two coincide exactly in the 57–68% of weeks (depending on W) in which the hold-out rule selects a single segment. Plain GDMA improves significantly on pooled weights for $W \geq 14$ and on per-BA weights for $W \leq 14$, but forcing all members of a segment onto identical weights costs 0.4–0.5% relative to soft segments when $W \geq 14$. Choosing G by the plain hold-out minimum instead of the one-standard-error rule is worse for short windows (0.823 against 0.816 at $W = 7$), confirming that parsimony matters when hold-out costs are noisy.

Fourth, the savings are economically large. At $W = 28$ the official-forecast policy costs 0.0761 units of idle-capacity cost per MWh of demand and soft GDMA 0.0609, a saving of 0.0152 units per MWh. Over the 4.0 billion MWh of annual demand in our panel, and valuing idle committed capacity at an illustrative \$10 per MWh, this is roughly \$600 million a year for the system as a whole, obtained by re-weighting forecasts that the operators already produce. The saving is not driven by a few operators: 80% of BAs are individually better off than under their own official forecast (last column of Table 3).

Table 4 and Figure 1 confirm the picture. With every BA weighted equally rather than

Table 4: Secondary metrics for $W = 28$ days.

Method	Scale-free cost ^b (rel. to EW)	Shortfall frequency	DM statistic vs soft GDMA ^c
Official forecast	0.7435	0.135	27.51
Equal weights	1.0000	0.083	35.98
Best single (per BA)	0.6326	0.138	7.66
Pooled DF weights	0.6262	0.117	5.34
Per-BA DF weights	0.6216	0.116	3.81
Shrinkage to pooled	0.6187	0.115	1.21
Two-step k -means	0.6218	0.120	2.90
Forecast-then-commit, per BA	0.6363	0.145	6.93
Forecast-then-commit, grouped	0.6346	0.137	7.29
GDMA	0.6244	0.119	4.53
GDMA, minimum hold-out rule	0.6225	0.120	4.62
Soft GDMA	0.6190	0.116	—
Target (mean τ_i across BAs)		0.125	

^b Each BA’s cost divided by its mean hourly demand, averaged over BAs with equal weight, relative to equal weights. ^c Diebold–Mariano statistic; positive values: the method has higher daily system cost than soft GDMA.

by its size, soft GDMA and shrinkage are again the two cheapest methods, so the results are not an artefact of the three largest ISOs. The decision-focused methods hit the target shortfall frequency closely (11.5–12.0% against a target of 12.5%, the average of $1 - \tau_i$), whereas the official-forecast policy and the forecast-then-commit methods fall short too often (13.5–14.5%). The cumulative saving builds up steadily over six years rather than in a few episodes, and the ordering of the methods is stable after the first year.

6.6 Segments

GDMA uses one segment in about two thirds of the weeks and two or more in the rest; multi-segment weeks occur in every year from 2021 onwards (Figure 2, left). The co-membership matrix (right) reveals two stable segments. The first contains 17 BAs, among them PJM, CAISO, ISO New England, Florida Power & Light, Arizona Public Service, Nevada Power and both PacifiCorp areas; their combinations rely almost entirely on the learned models (weights of 0.38 on the hourly regression and 0.47 on LightGBM) and put only 0.12 directly on the official forecast. The second contains 28 BAs, including MISO, ERCOT, the Southern Company, NYISO, TVA, Bonneville and Duke Energy, and puts 0.39 directly on the official forecast. Southwest Power Pool often forms a segment of its own. The two main segments have the same average critical ratio (0.874 and 0.873), so the segmentation is not a relabelling of cost asymmetries; nor does it follow geography or market design, since both contain ISOs and vertically integrated utilities from the East and the West. What separates them is how much the operator’s own forecast adds to what can be learned from recent demand—information that a central planner cannot read off the organisational chart of the grid, but that GDMA extracts from the data.

6.7 Extreme events and robustness

Table 5 examines three extreme episodes. During Winter Storm Uri and the 2023 heat wave, the decision-focused combinations cost 15–22% less than the official-forecast policy, in line with their

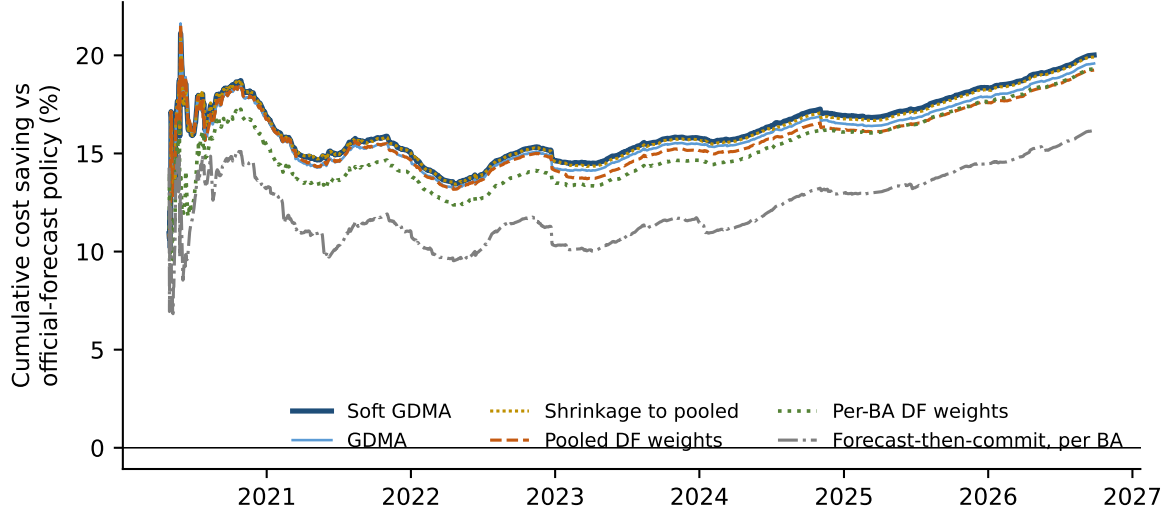


Figure 1: Cumulative system-wide cost saving relative to the official-forecast policy ($W = 28$).

Table 5: System cost during extreme events relative to the official-forecast policy ($W = 28$): Winter Storm Uri (10–20 February 2021), Winter Storm Elliott (22–27 December 2022) and the heat wave of 15 July–15 August 2023.

Method	Uri	Elliott	Heat wave 2023
Official forecast	1.000	1.000	1.000
Equal weights	1.133	2.534	1.366
Best single (per BA)	0.881	1.264	0.796
Pooled DF weights	0.842	1.318	0.778
Per-BA DF weights	0.848	1.184	0.790
Shrinkage to pooled	0.845	1.210	0.779
Two-step k -means	0.842	1.239	0.780
Forecast-then-commit, per BA	0.973	1.584	0.796
Forecast-then-commit, grouped	0.946	1.481	0.785
GDMA	0.842	1.318	0.782
GDMA, minimum hold-out rule	0.841	1.221	0.788
Soft GDMA	0.845	1.210	0.780

average performance. Winter Storm Elliott is different: the sudden cold snap of December 2022 was forecast much better by the operators, whose forecasts use weather information that none of our candidates sees directly, and every combination is 18–32% more expensive than the official policy over those six days. Soft GDMA and shrinkage limit the damage better than pooled weights and plain GDMA (1.21 against 1.32), presumably because unit-level shrinkage lets the BAs whose official forecasts are most informative keep more weight on them. The episode is a reminder that combination weights learned in calm weeks cannot anticipate an abrupt regime change; adding weather-based candidates, or letting weights respond to weather warnings, is a natural extension.

Table 6 replaces the calibrated critical ratios by a common τ for every BA, from the symmetric case $\tau = 0.5$ to the highly asymmetric $\tau = 0.95$. The level of the savings changes little—soft GDMA saves 19–23% relative to the official-forecast policy—and so does the ranking: soft GDMA is the cheapest method for $\tau \in \{0.5, 0.9, 0.95\}$ and under the calibrated ratios, and

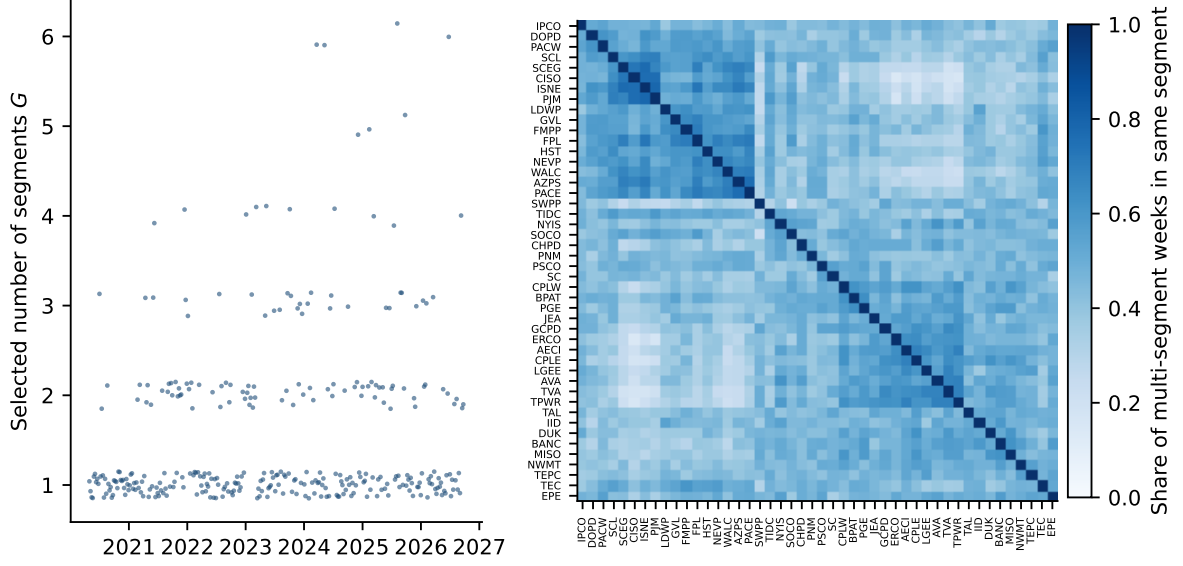


Figure 2: Left: number of segments selected by GDMA in each weekly re-estimation ($W = 28$). Right: among the weeks with $G \geq 2$, share of weeks in which two BAs belong to the same segment, ordered by average-linkage clustering.

ties with shrinkage for $\tau = 0.8$. The advantage of decision-focused over forecast-then-commit combination grows with the asymmetry, from 2.5% at $\tau = 0.5$ to 3.4% at $\tau = 0.95$ relative to the better forecast-then-commit method, as expected: the more asymmetric the cost, the more the best combination for the decision departs from the most accurate one.

7 Capacity commitment on five continents

7.1 A global panel of grid areas

We assemble public hourly demand data for 125 grid areas on five continents (Table 7). *North America* is represented by the 46 U.S. balancing authorities of Section 6 (EIA Form 930, 2019–2026), the only systems in the panel that publish their own day-ahead forecasts. *Europe* contributes 33 national systems from the Energy-Charts platform of Fraunhofer ISE (Fraunhofer Institute for Solar Energy Systems ISE, 2026), 2018–2026, including cross-border trading flows. *Oceania* contributes the five regions of Australia’s National Electricity Market (Australian Energy Market Operator, 2026), 2018–2026. *Asia* contributes the 31 provinces of mainland China for 2018 (Wu and Kan, 2023), the four grid areas of Taiwan for 2018 to mid-2022 (Fu and Chen, 2023) and the five regions of Thailand for 2023–2024 (Bunnak, 2025). *Africa* is represented by Algeria (Sonelgaz), 2018 to January 2020 (Rania, 2023). All series are converted to hourly UTC, cleaned as in Section 6, and placed on a common calendar from 1 January 2018 to 26 September 2026. Together the areas have an average demand of 1.66 TW—North America 455 GW, Europe 339 GW, Asia 839 GW, Oceania 21 GW and Africa 7 GW—roughly half of world electricity demand, although not all areas are observed at the same time.

The panel is deliberately unbalanced. Systems in Asia and Africa enter and leave, and several have only one or two years of data, which is typical of the data situation in emerging economies;

Table 6: Sensitivity to the cost asymmetry ($W = 28$): system cost relative to the official-forecast policy when every BA has the same critical ratio τ , and under the calibrated heterogeneous ratios.

Method	$\tau = 0.5$	$\tau = 0.8$	$\tau = 0.9$	$\tau = 0.95$	Calibrated
Official forecast	1.0000	1.0000	1.0000	1.0000	1.0000
Equal weights	1.4033	1.4353	1.4860	1.5265	1.5287
Best single (per BA)	0.8158	0.7921	0.8081	0.8326	0.8245
Pooled DF weights	0.7988	0.7775	0.7897	0.8136	0.8075
Per-BA DF weights	0.8007	0.7798	0.7914	0.8140	0.8069
Shrinkage to pooled	0.7932	0.7714	0.7834	0.8067	0.8009
Two-step k -means	0.7950	0.7736	0.7857	0.8101	0.8041
Forecast-then-commit, per BA	0.8136	0.7944	0.8144	0.8483	0.8382
Forecast-then-commit, grouped	0.8128	0.7916	0.8066	0.8337	0.8260
GDMA	0.7968	0.7750	0.7868	0.8094	0.8042
GDMA, minimum hold-out rule	0.7981	0.7762	0.7880	0.8098	0.8032
Soft GDMA	0.7931	0.7714	0.7832	0.8065	0.7999

Table 7: The global panel.

Continent	Systems	Areas	Period	Mean demand
North America	U.S. balancing authorities (EIA-930)	46	2019–2026	455 GW
Europe	National systems (Energy-Charts)	33	2018–2026	339 GW
Oceania	NEM regions, Australia (AEMO)	5	2018–2026	21 GW
Asia	China (31 provinces)	31	2018	
	Taiwan (4 grid areas)	4	2018–2022	839 GW
	Thailand (5 regions)	5	2023–2024	
Africa	Algeria (Sonelgaz)	1	2018–2020	7 GW
Total		125	2018–2026	1.66 TW

an area contributes to estimation and evaluation only on days with data. Public hourly data for other large African and Asian systems (South Africa, India, Japan, South Korea) exist but could not be retrieved without registration or were not reachable from our computing environment; we discuss this limitation in Section 9.

7.2 Candidates, costs and competitors

Every area receives nine candidate forecasts: the seven of Section 6; a zero-shot forecast from the pretrained time-series foundation model Chronos-Bolt (Ansari et al., 2024), which receives the previous 28 days of hourly demand and returns a 24-hour forecast without any fine-tuning; and a weather-aware global LightGBM model. The weather model adds to the LightGBM features the hourly 2-m temperature of the operating day, its daily mean, minimum and maximum, heating and cooling degrees (base 18 °C) and the previous day’s mean temperature. Temperatures come from NASA POWER (NASA Langley Research Center, 2026), which provides MERRA-2-based hourly data for any location without registration; each area is represented by the average of one to five load centres (169 locations in total, for example Dallas, Houston and San Antonio for ERCOT, or Paris, Lyon, Marseille and Lille for France). Realised temperature would be a perfect weather forecast, an *ex post* forecast in the terminology of Hong and Fan (2016). The weather model therefore uses, for the operating day, *archived day-ahead forecasts* of NOAA’s Global Forecast System (NOAA National Centers for Environmental Prediction, 2026): for

day d we take the run initialised at 00 UTC on day $d - 1$, which is available on the morning before the day-ahead commitment, and its 2-m temperature forecasts for lead times of 24 to 48 hours, interpolated to the load centres and to hourly resolution. Only the temperature field is downloaded from each 1-degree forecast file (18,846 fields, 99.8% retrieved). The public archive starts in January 2021, so the weather model has forecast temperatures from 2021 onwards and falls back on its other features before; the previous day’s temperature, which is known at the time of the decision, is always taken from NASA POWER. Against realised temperatures the GFS forecasts have a root-mean-squared error of 2.3°C (1.9°C in Europe, 2.2°C in North America, 2.7°C in Asia). A version of the model with realised temperatures is reported as an upper bound on the value of weather information. Over the evaluation period the mean absolute errors, scaled by mean demand, are 3.5% for the weather model (3.5% with exact temperatures), 3.6% for LightGBM, 3.8% for Chronos and for the hourly regression, 4.5% for the official/LightGBM candidate and 6–8% for the naive rules; the foundation model is thus competitive with a model trained on the data themselves.

Critical ratios follow the flexibility rule of Section 6 wherever interchange is measured (the United States, 31 European countries, and the isolated island grid of Taiwan, whose flexibility is zero); the remaining areas receive the midpoint $\tau = 0.875$. Section 6.7 and Table 18 in the appendix show that a common τ gives the same ranking.

Besides the benchmarks of Section 6 we add a deep-learning competitor, a *decision-focused neural gate*. It is a two-hidden-layer perceptron (6,920 parameters) that maps each area’s features at the weekly origin—each candidate’s recent decision cost and forecast error over the previous 7 and 28 days, the area’s critical ratio, size and continent, and the season—into combination weights through a softmax, as in mixture-of-experts model averaging (Gong et al., 2026). It is trained end to end on the same smoothed newsvendor cost, on the realised outcomes of the previous 52 weeks, and re-trained every four weeks. Unlike GDMA, it can let weights vary continuously with observed conditions; like GDMA, it only uses information available at the time of each decision.

7.3 Results

Table 8 shows that the results of the U.S. analysis hold on every continent. Worldwide, soft GDMA lowers the commitment cost by 18.0% relative to the single best forecaster, and its fair variant, discussed below, by 18.3%. Soft GDMA is significantly cheaper, at the 1% level, than every competitor except its own fair variant, which is cheaper still, and the two forecast-then-commit methods, which are 11% more expensive but whose daily costs are so volatile that the test is inconclusive; its margin over the closest competitors, shrinkage and two-step clustering, is small (0.5%) but statistically clear. One of the two GDMA variants has the lowest cost in Europe, Asia and Africa, and soft GDMA is tied with shrinkage in North America; in Oceania, with only five regions, the per-area, forecast-then-commit and neural-gate rules are 0.5–0.9% cheaper than fair soft GDMA.

Three comparisons stand out. First, decision-focused combination again dominates forecast-then-commit: combining the same forecasts for accuracy and adding a quantile margin saves only 9% worldwide and *raises* cost in Asia by 10–12%, where short histories make the estimated

Table 8: Out-of-sample commitment cost on five continents ($W = 28$ days, May 2018 to September 2026, nine candidates including the weather-aware model fed GFS forecasts), relative to the single best forecaster. Bold: lowest in the column. Stars: Diebold–Mariano test of equal daily cost against soft GDMA (Newey–West, 7 lags); *, **, ***: $p < 0.10, 0.05, 0.01$.

Method	World	North America	Europe	Oceania	Asia	Africa
Single best forecaster ^a	1.000***	1.000***	1.000***	1.000***	1.000***	1.000***
Equal weights	1.252***	1.383***	1.144***	1.000***	1.135***	1.022***
Best single (per area)	0.863***	0.843***	0.889***	0.815	0.891***	0.872***
Pooled DF weights	0.860***	0.835***	0.888***	0.871***	0.865***	0.825*
Per-area DF weights	0.829***	0.799	0.860***	0.798***	0.879***	0.830**
Shrinkage to pooled	0.824***	0.796	0.854***	0.812	0.851*	0.811
Two-step k -means	0.824***	0.801**	0.847*	0.812	0.852*	0.831**
Forecast-then-commit, per area	0.912*	0.927	0.870***	0.796***	1.124	0.840
Forecast-then-commit, grouped	0.907	0.940	0.844	0.799***	1.100	0.809
Neural gate (deep MoE)	0.916***	0.944***	0.882***	0.798***	0.985***	0.848***
GDMA	0.827***	0.808***	0.846*	0.827***	0.850*	0.813
Soft GDMA	0.820	0.797	0.843	0.813	0.848	0.808
Fair soft GDMA	0.817***	0.798	0.836***	0.803***	0.845	0.815*
Grid areas	125	46	33	5	40	1

^a The operator’s official forecast where one is published (United States), otherwise the global LightGBM forecast.

margins of the combined forecasts unreliable. Second, the neural gate is clearly worse than the simple decision-focused rules (0.916 against 0.820 worldwide) despite its flexibility; it is competitive only in Oceania. Its weights are learned from a year of past weeks and transfer poorly to weeks in which the relative performance of the candidates has shifted, whereas GDMA re-estimates a few interpretable rules every week. We did not tune the network extensively, so the comparison should be read as evidence that flexibility is not what this problem needs, not as a statement about the best achievable neural design. Third, better candidates and better combination are complements. The weather-aware model alone costs 0.943 of the single best forecaster, and adding it to the candidate set lowers the cost of soft GDMA from 0.828 to 0.820 (Table 9); combining remains 13% cheaper than the weather-aware model on its own. With realised temperatures instead of forecasts—a perfect weather forecast—every decision-focused and forecast-then-commit method becomes a further 0.8–1.0 points cheaper and the ranking is unchanged, so better weather forecasts would help all methods alike. Weather forecasts help only a little during Winter Storm Elliott: over those six days soft GDMA’s excess cost over the U.S. official forecasts falls from 7.7% without the weather candidate to 6.9% with the GFS-based candidate, and the best combination (two-step clustering) is still 3.3% more expensive than the operators. The operators’ forecasts use more weather information than 2-m temperature from one global model, and a sudden cold snap is exactly where that information matters. The ranking is stable across windows and critical ratios (Tables 17 and 18 in the appendix).

New and data-poor systems. Fifty-two areas enter the panel after the start of the evaluation: the 46 U.S. balancing authorities in January 2019, the five Thai regions in January 2023 and one European system. During their first eight weeks each of them has only a few weeks of history (Table 10). Pooled weights, which impose the rule learned from other systems, are 23% more expensive than the single best forecaster in this phase, and the neural gate, whose

Table 9: World commitment cost relative to the single best forecaster for three candidate sets ($W = 28$): eight candidates without weather, nine with the weather-aware model fed archived GFS day-ahead forecasts (main specification), and nine with realised temperatures (upper bound).

Method	No weather	GFS forecasts (main)	Realised weather
Single best forecaster ^a	1.000	1.000	1.000
Equal weights	1.315	1.252	1.250
Best single (per area)	0.870	0.863	0.858
Pooled DF weights	0.872	0.860	0.850
Per-area DF weights	0.838	0.829	0.821
Shrinkage to pooled	0.833	0.824	0.815
Two-step k -means	0.832	0.824	0.816
Forecast-then-commit, per area	0.923	0.912	0.904
Forecast-then-commit, grouped	0.917	0.907	0.898
Neural gate (deep MoE)	0.918	0.916	—
GDMA	0.835	0.827	0.818
Soft GDMA	0.828	0.820	0.812
Fair soft GDMA	—	0.817	0.807

Table 10: Commitment cost of areas entering the panel, relative to the single best forecaster ($W = 28$, nine candidates).

Method	First 8 weeks after entry ^b	All other weeks
Single best forecaster ^a	1.000	1.000
Equal weights	2.113	1.244
Best single (per area)	0.995	0.862
Pooled DF weights	1.232	0.856
Per-area DF weights	0.991	0.827
Shrinkage to pooled	0.993	0.822
Two-step k -means	0.967	0.822
Forecast-then-commit, per area	1.036	0.911
Forecast-then-commit, grouped	1.284	0.903
Neural gate (deep MoE)	1.762	0.908
GDMA	1.041	0.825
Soft GDMA	0.979	0.818
Fair soft GDMA	0.961	0.815
Grid areas entering	52	

^b The first eight weeks after an area has at least one day of data in the estimation window.

features are not yet informative, is 76% more expensive. Fair soft GDMA (0.961) and two-step clustering (0.967) are the cheapest methods for entering areas, followed by soft GDMA (0.978) and per-area weights (0.991): a new system borrows the rule of the segment it resembles after a few weeks, which is the mechanism behind the $\log G/T$ term of Theorem 1. Plain GDMA, which forces the new system onto its segment’s rule without unit-level shrinkage, is 4% more expensive than the single best forecaster in this phase—the clearest case for the soft version.

Segments across continents. GDMA uses more than one segment in 73% of the weekly re-estimations (two segments in 39%, three in 22%), much more often than in the U.S. panel, and soft GDMA shrinks towards the segment rule with an average intensity of $\kappa = 0.70$. Segments are not continents. Among weeks with several segments, two European systems share a segment 66% of the time and two U.S. balancing authorities 52% of the time, but European, Oceanian and African systems are grouped together 65–70% of the time, and North American areas are grouped with European ones only 28% of the time. The segmentation therefore does not follow

Table 11: Distribution of the benefits across the 125 grid areas ($W = 28$, nine candidates): percentiles of each area’s cost relative to its own single best forecaster, share of areas made worse off, median ratio in the worst-served continent, and cost ratios for data-rich and data-poor areas.

Method	Per-area cost ratio			Areas harmed	Worst continent ^d	Data-rich areas	Data-poor areas ^e
	P10	Median	P90				
Single best forecaster ^a	1.000	1.000	1.000	0.00	1.000	1.000	1.000
Equal weights	0.978	1.127	1.592	0.86	1.263	1.257	1.158
Best single (per area)	0.791	0.911	1.016	0.17	0.932	0.861	0.915
Pooled DF weights	0.790	0.892	1.014	0.13	0.902	0.860	0.862
Per-area DF weights	0.775	0.882	0.984	0.06	0.910	0.825	0.903
Shrinkage to pooled	0.775	0.867	0.979	0.05	0.880	0.822	0.863
Two-step k -means	0.781	0.872	0.973	0.04	0.883	0.821	0.866
Forecast-then-commit, per area	0.795	0.915	1.156	0.27	0.967	0.912	0.916
Forecast-then-commit, grouped	0.793	0.895	1.157	0.26	0.974	0.908	0.871
Neural gate (deep MoE)	0.805	0.918	1.128	0.33	1.035	0.910	1.025
GDMA	0.782	0.868	0.985	0.08	0.885	0.825	0.863
Soft GDMA	0.774	0.862	0.971	0.04	0.875	0.818	0.863
Fair soft GDMA	0.772	0.855	0.957	0.02	0.869	0.815	0.859
Areas	125					93	32

^d Highest continent-level median of the per-area ratios. ^e Areas with less than two years of evaluation data (the 31 Chinese provinces and Algeria).

geography; it is consistent with differences in which forecasting systems work—notably whether an operator’s own forecast is available and informative—which a planner could not have specified in advance.

7.4 Who benefits? A fairness audit

Table 11 asks how the gains are distributed. Methods that impose a common rule or learn a flexible global map spread their gains unevenly: pooled weights make 13% of the areas worse off than their own single best forecaster, and the neural gate and the forecast-then-commit methods 26–33%; for the neural gate the median area in the worst-served continent loses 3.5%. Soft GDMA makes only 4% of areas worse off and has the best worst-served continent among the efficiency-oriented methods, and fair soft GDMA improves on every count (2% harmed, the tightest distribution with a 90th percentile of 0.957, and a worst-served continent at 0.869). A gap remains between data-rich and data-poor systems: data-poor areas save 14% under the GDMA variants against 18–19% for data-rich ones, and per-area weights, which have little data to work with for these systems, save only 10%.

Strikingly, fairness costs nothing here. Fair soft GDMA, which counts every area’s relative improvement equally, is also the cheapest method in total (0.817 against 0.820, significant at the 1% level). Weighting by relative cost prevents the few largest systems from dictating the segment rules, and the rules that serve the many smaller systems well turn out to generalise better out of sample—consistent with the segment-structure argument of Section 4, under which fair and efficient criteria share their minimiser.

Income. Fairness at this scale is also a question of income. We attach to every grid area the GDP per capita at purchasing-power parity of its country and its World Bank income group (World Bank, 2026), with Taiwan from the IMF World Economic Outlook (International

Table 12: Benefits by income ($W = 28$): commitment cost relative to the single best forecaster by World Bank income group and, within the United States, for the balancing authorities in the lowest and highest third of state per-capita income; Spearman rank correlations between an area’s cost ratio and income.

Method	All 125 areas			U.S. balancing authorities		
	High income	Upper-middle income	Rank corr. with GDP p.c. ^g	Lowest income third	Highest income third	Rank corr. with state income ^g
Single best forecaster ^a	1.000	1.000	—	1.000	1.000	—
Equal weights	1.261	1.123	0.197	1.597	1.084	-0.131
Best single (per area)	0.860	0.906	-0.087	0.944	0.667	-0.006
Pooled DF weights	0.859	0.869	0.030	0.934	0.689	-0.131
Per-area DF weights	0.824	0.893	-0.199	0.894	0.629	-0.091
Shrinkage to pooled	0.821	0.863	-0.092	0.890	0.636	-0.099
Two-step k -means	0.821	0.865	-0.066	0.893	0.651	-0.044
Forecast-then-commit, per area	0.910	0.941	-0.056	1.043	0.716	-0.161
Forecast-then-commit, grouped	0.907	0.908	0.039	1.060	0.747	-0.171
Neural gate (deep MoE)	0.911	0.980	-0.241	1.064	0.740	-0.173
GDMA	0.824	0.864	-0.034	0.901	0.659	-0.063
Soft GDMA	0.817	0.861	-0.073	0.890	0.642	-0.071
Fair soft GDMA	0.814	0.857	-0.070	0.885	0.643	-0.082
Areas	83	42		15	16	

^g A negative correlation means that richer areas have lower cost ratios, i.e. gain more.

Monetary Fund, 2026); 83 areas are in high-income and 42 in upper-middle-income countries. For the U.S. balancing authorities we also compute the per-capita personal income of the states they serve (U.S. Bureau of Economic Analysis, 2026), weighted by approximate service-territory shares. Table 12 reports the results.

Across the world, areas in upper-middle-income countries save less than areas in high-income countries under every decision-focused method (soft GDMA 13.9% against 18.3%), but this gap largely coincides with the data gap above, because the upper-middle-income group is dominated by the Chinese provinces with a single year of data. Across individual areas, the rank correlation between an area’s cost ratio and its GDP per capita is close to zero for the GDMA variants (-0.03 to -0.07) and for pooled weights, and is most negative—richer areas gaining more—for the neural gate (-0.24) and per-area weights (-0.20). Within the United States the picture is similar. In cost-weighted terms the balancing authorities in the highest third of state income gain more (cost ratios of 0.64–0.66 under the GDMA variants against 0.89–0.90 in the lowest third), but this reflects a few large, high-income systems whose own official forecasts happen to be weak; across balancing authorities the rank correlation with state income is again close to zero (-0.06 to -0.08). We find no evidence that GDMA systematically favours richer systems, and its fair variant slightly narrows every income gap in the table.

8 Where to install batteries

We now solve the planning problem (4) on the global panel. For each commitment rule we take its out-of-sample hourly shortfalls and surpluses ($W = 28$, nine candidates with GFS weather forecasts), place a four-hour battery with 85% round-trip efficiency in every grid area, and compute the annual value of battery power on a grid from 0 to 15% of the area’s mean demand. Costs are converted to money by valuing one MWh of idle committed capacity at \$10, so that a shortfall costs between \$40 and \$190 per MWh depending on the area’s critical ratio; the value of battery power scales proportionally with this assumption.

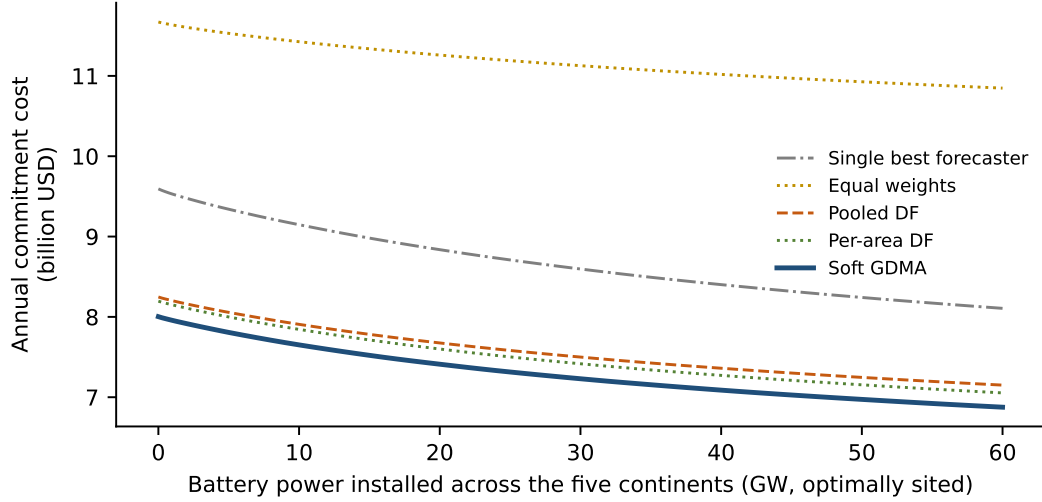


Figure 3: Annual system-wide commitment cost as a function of battery power installed and optimally sited across the 125 grid areas, for five commitment rules. Costs valued at \$10 per MWh of idle committed capacity.

Table 13: Storage siting on five continents. Cost of each commitment rule without storage, battery power it needs (optimally sited) to reach the cost of soft GDMA without storage, and the optimal siting of a 10 GW storage programme.

Commitment rule	Cost without storage (\$bn/yr)	GW to match soft GDMA ^c	Siting of a 10 GW budget (GW)				
			N. Am.	Europe	Oceania	Asia	Africa
Single best forecaster	9.59	> 60	3.7	1.9	0.4	4.0	0.0
Equal weights	11.67	> 60	1.4	2.0	1.0	5.5	0.1
Pooled DF	8.25	7	2.1	2.4	0.5	5.0	0.0
Per-area DF	8.20	5	3.1	2.6	0.5	3.7	0.0
Forecast-then-commit	8.30	7	4.2	2.5	0.4	2.8	0.1
GDMA	8.05	1	2.5	2.5	0.5	4.4	0.1
Soft GDMA	8.00	–	2.6	2.5	0.5	4.4	0.0

^c Smallest optimally sited budget, on a grid of 0.25 GW steps up to 5 GW and 1 GW steps up to 60 GW, at which the rule's cost falls to or below that of soft GDMA without storage.

8.1 Batteries do not pay for themselves on forecast errors alone

The first increment of battery power is worth \$37,000–\$46,000 per MW-year in the median area of each continent (depending on the commitment rule), and at most \$84,000 (in Great Britain). Annualised costs of four-hour batteries are well above this range (we use \$100,000–\$200,000 per MW-year), so the cost-optimal battery size for covering day-ahead commitment errors alone is zero in every area and for every commitment rule. Storage is deployed for arbitrage, capacity adequacy and the integration of renewables, and covering commitment errors is a by-product. The relevant planning question is therefore not whether, but *where*, to place the storage that energy-transition programmes already mandate, and how that choice depends on the way operators commit.

Table 14: Annual value (million USD) of storage programmes of different sizes when operations follow soft GDMA, by the commitment rule whose error profile was used to site the batteries.

Siting based on	1 GW	5 GW	10 GW	20 GW
Single best forecaster	39	182	342	581
Per-area DF	45	193	350	590
Forecast-then-commit	40	186	342	585
Soft GDMA	46	196	352	592

8.2 Better forecasting substitutes for storage

Figure 3 and Table 13 give the central managerial result. Without storage, soft GDMA commits the five continents’ capacity for \$8.00 billion a year, against \$9.59 billion for the single best forecaster. Adding batteries lowers every curve, but the curve of the single best forecaster never reaches soft GDMA’s starting point: even 60 GW of optimally sited batteries leave it at \$8.11 billion. Relative to the standard ways of combining forecasts, soft GDMA is worth 5 GW (per-area weights) and 7 GW (forecast-then-commit and pooled weights) of batteries. At \$150,000 per MW-year, 5–7 GW of batteries cost \$0.75–1.05 billion a year; the re-weighting of existing forecasts that replaces them costs essentially nothing.

8.3 The siting decision depends on the commitment rule

Table 13 also shows how a 10 GW programme should be distributed. Under soft GDMA, 2.6 GW go to North America, 2.5 GW to Europe, 0.5 GW to Oceania and 4.4 GW to Asia, and Africa receives almost nothing. If the planner instead uses the error profile of the single best forecaster, North America receives 3.7 GW, about 40% more, because the status quo leaves U.S. operators with the largest shortfalls; the forecast-then-commit rule would send 4.2 GW there. Those batteries would be in the wrong place once operations improve.

Table 14 quantifies the cost of that mistake. When operations follow soft GDMA, a 1 GW programme sited with the status-quo error profile delivers \$39 million a year instead of \$46 million, a 15% loss; the loss is 7% at 5 GW, 3% at 10 GW and 2% at 20 GW, because large programmes eventually reach every area with sizeable shortfalls. Small, early programmes—the typical first phase of a national storage strategy—are therefore the most sensitive to the operational rule assumed at the planning stage. The implication for planners is sequencing: fix how forecasts are combined first, and site storage on the residual risk that remains.

8.4 Fair siting

Efficient siting concentrates batteries where the value in money is highest, and in small programmes some continents receive nothing: with 1 GW, Africa’s relative benefit is zero (Table 15). Two fair rules give different answers. Floors proportional to each continent’s share of demand, the “fair share” rule often used in practice, cost 0.7–1.5% of the programme’s value and *lower* the relative benefit of the worst-served continent at 5 GW and above, because they move batteries from small systems with large shortfalls, such as those in Oceania, to large systems where each MW achieves little in relative terms. Maximin siting, which directs every increment to the continent that has gained least so far, raises the worst-served continent’s relative benefit

Table 15: Efficient and fair siting of storage programmes under soft GDMA operations: annual value, the relative benefit of the worst-served continent, and the price of fairness (PoF, the share of the efficient programme’s value given up).

Budget (GW)	Efficient		Proportional floors ($\lambda = 1$)			Maximin		
	Value (\$m/yr)	Min. rel. benefit ^f	Value (\$m/yr)	Min. rel. benefit	PoF	Value (\$m/yr)	Min. rel. benefit	PoF
1	46	0.0%	46	0.3%	0.7%	46	0.5%	1.2%
5	196	2.2%	194	1.4%	1.0%	195	2.4%	0.4%
10	352	3.8%	347	2.7%	1.2%	350	4.3%	0.5%
20	592	6.6%	583	5.1%	1.5%	588	7.3%	0.6%

^f Lowest, across continents, of the value captured divided by the continent’s annual commitment cost without storage.

at every budget—for example from 3.8% to 4.3% of its commitment cost with 10 GW, which doubles Africa’s allocation, and from zero to 0.5% with 1 GW—and gives up only 0.4–1.2% of the programme’s value. For planners the lesson is that fairness in storage siting is cheap if it is defined by who benefits, and can be counter-productive if it is defined by who consumes.

9 Managerial implications and conclusion

Implications for system operators and planners. Five lessons emerge for organisations that coordinate many operating units, from grid operators and regulators to international bodies that fund energy-transition programmes. First, the object to be learned is the *decision rule*, not the forecast: the cost-minimising combination of forecasting systems depends on the cost asymmetry each unit faces, and combining for accuracy leaves 3–16% of the achievable saving on the table, and in data-poor systems can even raise cost. Second, planners need not choose between one rule for every system and a bespoke rule for each. A handful of data-driven segments, with each unit shrunk towards its segment’s rule, captures the heterogeneity that matters, is estimated with far less noise than bespoke rules, and produces a segmentation that can be discussed with operators—who is grouped with whom, and why. Third, more flexible is not better: a neural mixture-of-experts gate with thousands of parameters, trained on the same decision loss, was ten percentage points more expensive worldwide than soft GDMA, much more expensive for systems with little history, and made a third of the grid areas worse off. Fourth, fairness need not cost efficiency: counting every area’s relative improvement equally produced rules that were both cheaper in total and more evenly beneficial, and in storage siting a maximin rule protected the worst-served continent at a price of fairness of about 1%, whereas the intuitive demand-proportional “fair share” made the worst-served continent worse off. Fifth, forecasting and investment are linked. Better commitment rules are worth several gigawatts of batteries, and a storage programme should be sited on the risk that remains after operations have been improved; siting it on today’s error profile misplaces a meaningful share of a small programme’s value. Because GDMA only re-weights forecasts that operators already produce, it can be adopted without replacing existing forecasting systems.

Limitations and extensions. Our coverage of Africa and parts of Asia is thin, because public hourly demand data for South Africa, India, Japan and South Korea could not be retrieved in our computing environment; the method is designed for exactly such data-poor systems, the audit shows that they still benefit less than data-rich ones, and extending the panel is a priority. The fairness analysis measures income at the level of countries and U.S. states; the distribution of benefits across the households served within an area would require data we do not have. The cost calibration is stylised: shortfall costs are tied to a transparent proxy of import flexibility rather than to market prices, and the single-period newsvendor abstracts from ramping, network constraints and multi-day commitments. The weather candidate uses archived GFS temperature forecasts, which are public only from 2021 and cover a single variable from a single global model; the operators’ own forecasts, which draw on richer weather information, remain better during the most abrupt event in our sample, Winter Storm Elliott, and adding wind, humidity and higher-resolution or ensemble forecasts is a natural extension. The storage stage takes the commitment rule as given; jointly optimising commitments and battery dispatch, and adding the arbitrage and adequacy value of storage, would make the siting problem richer. The theory treats the number of segments as given and relies on a high-level concentration condition; a formal analysis of the hold-out choice of G and κ , and of strong cross-sectional dependence through common weather shocks, is left for future work.

Conclusion. Grouped decision-focused model averaging combines three ideas that have so far developed separately—optimal model averaging, decision-focused learning and latent group structures—into one estimator with finite-sample and asymptotic guarantees, and links it to the siting of new capacity. On five continents it turns the forecasts operators already have into commitment rules that are cheaper, statistically stable and interpretable, and it tells planners where new storage is worth most once those rules are in place.

References

- Aiolfi, M. and Timmermann, A. (2006). Persistence in forecasting performance and conditional combination strategies. *Journal of Econometrics*, 135(1–2):31–53.
- Ando, T. and Bai, J. (2016). Panel data models with grouped factor structure under unknown group membership. *Journal of Applied Econometrics*, 31(1):163–191.
- Ansari, A. F., Stella, L., Turkmen, C., Zhang, X., Mercado, P., Shen, H., Shchur, O., Ranganapuram, S. S., Arango, S. P., Kapoor, S., Zschiegner, J., Maddix, D. C., Wang, H., Mahoney, M. W., Torkkola, K., Wilson, A. G., Bohlke-Schneider, M., and Wang, Y. (2024). Chronos: Learning the language of time series. *Transactions on Machine Learning Research*.
- Arthur, D. and Vassilvitskii, S. (2007). k-means++: The advantages of careful seeding. In *Proceedings of the Eighteenth Annual ACM-SIAM Symposium on Discrete Algorithms*, pages 1027–1035.
- Australian Energy Market Operator (2026). Aggregated price and demand data, National Electricity Market. <https://www.aemo.com.au/energy-systems/electricity/>

- [national-electricity-market-nem/data-nem/aggregated-data](#), accessed 29 September 2026.
- Ban, G.-Y. and Rudin, C. (2019). The big data newsvendor: Practical insights from machine learning. *Operations Research*, 67(1):90–108.
- Bates, J. M. and Granger, C. W. J. (1969). The combination of forecasts. *Operational Research Quarterly*, 20(4):451–468.
- Bertsimas, D., Farias, V. F., and Trichakis, N. (2011). The price of fairness. *Operations Research*, 59(1):17–31.
- Bertsimas, D. and Kallus, N. (2020). From predictive to prescriptive analytics. *Management Science*, 66(3):1025–1044.
- Bonhomme, S. and Manresa, E. (2015). Grouped patterns of heterogeneity in panel data. *Econometrica*, 83(3):1147–1184.
- Bunnak, P. (2025). Thai power system: hourly power generation, demand, and cross-border flows. Zenodo, <https://zenodo.org/records/17109911>.
- Busby, J. W., Baker, K., Bazilian, M. D., Gilbert, A. Q., Grubert, E., Rai, V., Rhodes, J. D., Shidore, S., Smith, C. A., and Webber, M. E. (2021). Cascading risks: Understanding the 2021 winter blackout in Texas. *Energy Research & Social Science*, 77:102106.
- Claeskens, G., Magnus, J. R., Vasnev, A. L., and Wang, W. (2016). The forecast combination puzzle: A simple theoretical explanation. *International Journal of Forecasting*, 32(3):754–762.
- Diebold, F. X. and Shin, M. (2019). Machine learning for regularized survey forecast combination: Partially-egalitarian LASSO and its derivatives. *International Journal of Forecasting*, 35(4):1679–1691.
- Elmachtoub, A. N. and Grigas, P. (2022). Smart “predict, then optimize”. *Management Science*, 68(1):9–26.
- Fraunhofer Institute for Solar Energy Systems ISE (2026). Energy-charts public API: national electricity load and cross-border trading. <https://api.energy-charts.info>, accessed 29 September 2026.
- Fu, C. and Chen, B. (2023). Taiwan electricity generation and demand data, 2017 January–2022 July. Zenodo, <https://zenodo.org/records/7537890>.
- Gneiting, T. (2011). Quantiles as optimal point forecasts. *International Journal of Forecasting*, 27(2):197–207.
- Gong, S., He, B., and Zhang, X. (2026). Optimal mixture-of-experts model averaging for conditional generative models. arXiv:2607.04360.
- Granger, C. W. J. (1969). Prediction with a generalized cost of error function. *Operational Research Quarterly*, 20(2):199–207.

- Hansen, B. E. (2007). Least squares model averaging. *Econometrica*, 75(4):1175–1189.
- Hansen, B. E. and Racine, J. S. (2012). Jackknife model averaging. *Journal of Econometrics*, 167(1):38–46.
- Hastie, T., Tibshirani, R., and Friedman, J. (2009). *The Elements of Statistical Learning*. Springer, New York, 2nd edition.
- Hong, T. and Fan, S. (2016). Probabilistic electric load forecasting: A tutorial review. *International Journal of Forecasting*, 32(3):914–938.
- Hubert, L. and Arabie, P. (1985). Comparing partitions. *Journal of Classification*, 2:193–218.
- International Monetary Fund (2026). World economic outlook database: GDP per capita, current prices, purchasing power parity. <https://www.imf.org/external/datamapper>, accessed 29 September 2026.
- Ke, G., Meng, Q., Finley, T., Wang, T., Chen, W., Ma, W., Ye, Q., and Liu, T.-Y. (2017). LightGBM: A highly efficient gradient boosting decision tree. In *Advances in Neural Information Processing Systems*, volume 30.
- Larroux, G., Jacobs, M., Jia, K., Sossan, F., and Paolone, M. (2026). Enhancing power systems transmission adequacy via optimal BESS siting and sizing using Benders decomposition with feasibility cuts. arXiv:2603.04189.
- Levi, R., Perakis, G., and Uichanco, J. (2015). The data-driven newsvendor problem: New bounds and insights. *Operations Research*, 63(6):1294–1306.
- Lin, C.-C. and Ng, S. (2012). Estimation of panel data models with parameter heterogeneity when group membership is unknown. *Journal of Econometric Methods*, 1(1):42–55.
- Lu, X. and Su, L. (2015). Jackknife model averaging for quantile regressions. *Journal of Econometrics*, 188(1):40–58.
- Mandi, J., Kotary, J., Berden, S., Mulamba, M., Bucarey, V., Guns, T., and Fioretto, F. (2024). Decision-focused learning: Foundations, state of the art, benchmark and future opportunities. *Journal of Artificial Intelligence Research*, 80:1623–1701.
- Merlevède, F., Peligrad, M., and Rio, E. (2011). A Bernstein type inequality and moderate deviations for weakly dependent sequences. *Probability Theory and Related Fields*, 151:435–474.
- Montero-Manso, P., Athanasopoulos, G., Hyndman, R. J., and Talagala, T. S. (2020). FFORMA: Feature-based forecast model averaging. *International Journal of Forecasting*, 36(1):86–92.
- NASA Langley Research Center (2026). POWER project: hourly meteorological data (MERRA-2 based), API version 2. <https://power.larc.nasa.gov>, accessed 29 September 2026.

- NOAA National Centers for Environmental Prediction (2026). Global forecast system (GFS) 1-degree forecasts, NOAA open data dissemination archive (noaa-gfs-bdp-pds). <https://noaa-gfs-bdp-pds.s3.amazonaws.com>, accessed 29 September 2026.
- Piansky, R., Stinchfield, G., Kody, A., Molzahn, D. K., and Watson, J.-P. (2024). Long duration battery sizing, siting, and operation under wildfire risk using progressive hedging. arXiv:2404.12296.
- Qu, Z., Wang, W., and Zhang, X. (2025). Prediction intervals for model averaging. arXiv:2510.16224.
- Rania, F. (2023). Load consumption data Algeria. Mendeley Data, version 1, doi:10.17632/z5x2d3mhw7.1.
- Ruggles, T. H., Farnham, D. J., Tong, D., and Caldeira, K. (2020). Developing reliable hourly electricity demand data through screening and imputation. *Scientific Data*, 7:155.
- Sagawa, S., Koh, P. W., Hashimoto, T. B., and Liang, P. (2020). Distributionally robust neural networks for group shifts: On the importance of regularization for worst-case generalization. In *International Conference on Learning Representations*.
- Smith, J. and Wallis, K. F. (2009). A simple explanation of the forecast combination puzzle. *Oxford Bulletin of Economics and Statistics*, 71(3):331–355.
- Sovacool, B. K. and Dworkin, M. H. (2015). Energy justice: Conceptual insights and practical applications. *Applied Energy*, 142:435–444.
- Su, L., Shi, Z., and Phillips, P. C. B. (2016). Identifying latent structures in panel data. *Econometrica*, 84(6):2215–2264.
- Sun, Y., Hong, Y., Lee, T.-H., Wang, S., and Zhang, X. (2021). Time-varying model averaging. *Journal of Econometrics*, 222(2):974–992.
- Timmermann, A. (2006). Forecast combinations. In Elliott, G., Granger, C. W. J., and Timmermann, A., editors, *Handbook of Economic Forecasting*, volume 1, pages 135–196. Elsevier, Amsterdam.
- U.S. Bureau of Economic Analysis (2026). Per capita personal income by state, retrieved from FRED, Federal Reserve Bank of St. Louis. <https://fred.stlouisfed.org>, series <ST>PCPI, accessed 29 September 2026.
- U.S. Energy Information Administration (2026). Hourly electric grid monitor (Form EIA-930): bulk data file EBA.zip. <https://www.eia.gov/opensdata/bulk/EBA.zip>, accessed 29 September 2026.
- Vershynin, R. (2018). *High-Dimensional Probability: An Introduction with Applications in Data Science*. Cambridge University Press, Cambridge.
- Wang, X., Hyndman, R. J., Li, F., and Kang, Y. (2023). Forecast combinations: An over 50-year review. *International Journal of Forecasting*, 39(4):1518–1547.

- Wang, Y., Zhang, N., Tan, Y., Hong, T., Kirschen, D. S., and Kang, C. (2019). Combining probabilistic load forecasts. *IEEE Transactions on Smart Grid*, 10(4):3664–3674.
- World Bank (2026). World development indicators: GDP per capita, PPP (current international \$), NY.GDP.PCAP.PP.CD, and country income classification. <https://api.worldbank.org/v2>, accessed 29 September 2026.
- Wu, H. and Kan, X. (2023). Hourly electric power load and transmission data at the provincial level in China. Zenodo, <https://zenodo.org/records/8322210>.
- Yu, B. (1994). Rates of convergence for empirical processes of stationary mixing sequences. *Annals of Probability*, 22(1):94–116.
- Yu, D., Zhang, X., and Liang, H. (2024). Unified optimal model averaging with a general loss function based on cross-validation. arXiv:2412.09804.
- Zhang, X. and Liu, C.-A. (2023). Model averaging prediction by K-fold cross-validation. *Journal of Econometrics*, 235(1):280–301.
- Zhang, X., Wan, A. T. K., and Zou, G. (2013). Model averaging by jackknife criterion in models with dependent data. *Journal of Econometrics*, 174(2):82–94.
- Zhang, X. and Zhang, X. (2023). Optimal model averaging based on forward-validation. *Journal of Econometrics*, 237(2).
- Zhang, Y., Xie, T., and Zhang, X. (2026). Handling covariate shift by model averaging. arXiv:2608.14025.
- Zou, J., Vasnev, A., Wang, W., and Zhang, X. (2025). A theoretical comparison of weight constraints in forecast combination and model averaging. arXiv:2510.26456.

A Proofs

Throughout, $\|\cdot\|_1$ is the ℓ_1 norm and constants c, C may change from line to line.

Covering the simplex. For $\epsilon \in (0, 1]$ there is a set $\mathcal{N}_\epsilon \subset \Delta_M$ with $|\mathcal{N}_\epsilon| \leq (3/\epsilon)^M$ such that every $\mathbf{w} \in \Delta_M$ has a point $\mathbf{w}' \in \mathcal{N}_\epsilon$ with $\|\mathbf{w} - \mathbf{w}'\|_1 \leq \epsilon$ (volumetric argument for the ℓ_1 ball, e.g. Vershynin, 2018, Ch. 4). Hence $\mathcal{N}_\epsilon^G$ is an ϵ -net of Δ_M^G in the norm $\max_g \|\mathbf{w}_g - \mathbf{w}'_g\|_1$, of cardinality at most $(3/\epsilon)^{GM}$.

Lipschitz property. For all $(\gamma, \mathbf{W})$ and $(\gamma, \mathbf{W}')$ with $\max_g \|\mathbf{w}_g - \mathbf{w}'_g\|_1 \leq \epsilon$, Assumption 1 gives $|\hat{R}(\gamma, \mathbf{W}) - \hat{R}(\gamma, \mathbf{W}')| \leq \bar{a}B\epsilon$ and $|R(\gamma, \mathbf{W}) - R(\gamma, \mathbf{W}')| \leq \bar{a}B\epsilon$.

Proof of Theorem 1

Fix $\epsilon = 1/(NT_e)$. The set $\mathcal{F} = \{1, \dots, G\}^N \times \mathcal{N}_\epsilon^G$ has cardinality at most $G^N(3NT_e)^{GM}$. By Assumption 2 and the union bound,

$$\Pr \left(\max_{(\gamma, \mathbf{W}) \in \mathcal{F}} |\widehat{R}(\gamma, \mathbf{W}) - R(\gamma, \mathbf{W})| > x \right) \leq 2G^N(3NT_e)^{GM} \exp(-cNT_e x^2/L^2).$$

Setting the right-hand side equal to δ gives, with probability at least $1 - \delta$, $\max_{\mathcal{F}} |\widehat{R} - R| \leq x_\delta := L\{(GM \log(3NT_e) + N \log G + \log(2/\delta))/(cNT_e)\}^{1/2}$. For an arbitrary $(\gamma, \mathbf{W}) \in \mathcal{P}_G$ let $\mathbf{W}'$ be its nearest net point; by the Lipschitz property,

$$\sup_{\mathcal{P}_G} |\widehat{R} - R| \leq x_\delta + 2\bar{a}B\epsilon.$$

Let $(\gamma^*, \mathbf{W}^*)$ satisfy $R(\gamma^*, \mathbf{W}^*) \leq R_G^* + \eta$ for arbitrary $\eta > 0$. Since $(\widehat{\gamma}, \widehat{\mathbf{W}})$ minimises $\widehat{R}$,

$$R(\widehat{\gamma}, \widehat{\mathbf{W}}) \leq \widehat{R}(\widehat{\gamma}, \widehat{\mathbf{W}}) + x_\delta + 2\bar{a}B\epsilon \leq \widehat{R}(\gamma^*, \mathbf{W}^*) + x_\delta + 2\bar{a}B\epsilon \leq R_G^* + \eta + 2x_\delta + 4\bar{a}B\epsilon.$$

Letting $\eta \downarrow 0$ and substituting $\epsilon = 1/(NT_e)$ gives (5). $\square$

Proof of Corollary 1

Theorem 1 with $\delta = \delta_{N,T} \rightarrow 0$ slowly (for instance $\log(1/\delta) = \log(NT_e)$) shows $R(\widehat{\gamma}, \widehat{\mathbf{W}}) - R_G^* = O_p(r_{N,T})$ with $r_{N,T} = \{GM \log(NT_e)/(NT_e) + \log G/T_e + \log(NT_e)/(NT_e)\}^{1/2} \rightarrow 0$. Dividing by $R_G^* \geq \kappa$ yields $R(\widehat{\gamma}, \widehat{\mathbf{W}})/R_G^* - 1 = O_p(r_{N,T}/\kappa) \rightarrow_p 0$. Because $R(\widehat{\gamma}, \widehat{\mathbf{W}}) \geq R_N^*$ always and $R_G^* \geq R_N^*$, the second claim follows once we show $R_G^* = R_N^*$ under the stated structure. Let $\mathbf{w}_g^0$ be a common minimiser of R_i for all i with $\gamma_i^0 = g$ (padding with arbitrary vectors if $G_0 < G$); then $R(\gamma^0, \mathbf{W}^0) = N^{-1} \sum_i \min_{\mathbf{w}} R_i(\mathbf{w}) = R_N^*$, so $R_G^* \leq R_N^*$, and the reverse inequality holds because $\mathcal{P}_G$ is a subset of the unit-specific class. $\square$

Proof of Theorem 2

Step 1 (average consistency). Under Assumption 3(a), $R_G^* = R(\gamma^0, \mathbf{W}^0) = R_N^*$ and

$$\frac{\lambda}{N} \sum_{i=1}^N \|\widehat{\mathbf{w}}_{\widehat{\gamma}_i} - \mathbf{w}_{\gamma_i^0}^0\|_2^2 \leq R(\widehat{\gamma}, \widehat{\mathbf{W}}) - R_G^* = o_p(1)$$

by Theorem 1, since $\log N/T_e \rightarrow 0$ implies $\log G/T_e \rightarrow 0$ and $GM \log(NT_e)/(NT_e) \rightarrow 0$ for fixed G and M .

Step 2 (consistency of the segment weights). Fix $\eta \in (0, d/4)$. By Step 1 and Markov's inequality, the fraction of units with $\|\widehat{\mathbf{w}}_{\widehat{\gamma}_i} - \mathbf{w}_{\gamma_i^0}^0\|_2 > \eta$ is $o_p(1)$. Since every true segment g contains at least πN units, with probability approaching one each g contains a unit i with $\|\widehat{\mathbf{w}}_{\widehat{\gamma}_i} - \mathbf{w}_g^0\|_2 \leq \eta$; call $h(g) = \widehat{\gamma}_i$. If $h(g) = h(g')$ for $g \neq g'$, the triangle inequality gives $\|\mathbf{w}_g^0 - \mathbf{w}_{g'}^0\|_2 \leq 2\eta < d$, contradicting Assumption 3(b). Thus h is injective, hence a bijection of $\{1, \dots, G\}$, and after relabelling $\max_g \|\widehat{\mathbf{w}}_g - \mathbf{w}_g^0\|_2 \leq \eta$ with probability approaching one. Since η is arbitrary, (i) follows.

Step 3 (memberships). On the event $E_\eta = \{\max_g \|\hat{\mathbf{w}}_g - \mathbf{w}_g^0\|_2 \leq \eta\}$, unit i with $\gamma_i^0 = g$ can be assigned to $h \neq g$ only if $\hat{R}_i(\hat{\mathbf{w}}_h) \leq \hat{R}_i(\hat{\mathbf{w}}_g)$; every global minimiser assigns each unit to a segment minimising its own empirical cost, since otherwise reassigning the unit would lower (2). By Assumption 3 and the Lipschitz property (with $\|\cdot\|_1 \leq \sqrt{M}\|\cdot\|_2$),

$$R_i(\hat{\mathbf{w}}_h) - R_i(\hat{\mathbf{w}}_g) \geq \lambda(d - \eta)^2 - \bar{a}B\sqrt{M}\eta \geq \lambda d^2/4 =: \Delta$$

for η small enough. Hence misclassification of unit i on E_η requires $\sup_{\mathbf{w} \in \Delta_M} |\hat{R}_i(\mathbf{w}) - R_i(\mathbf{w})| \geq \Delta/2$. Covering Δ_M by $\mathcal{N}_\epsilon$ with $\epsilon = \Delta/(8\bar{a}B)$ and applying the unit-level part of Assumption 2,

$$\Pr\left(\sup_{\mathbf{w}} |\hat{R}_i(\mathbf{w}) - R_i(\mathbf{w})| \geq \Delta/2\right) \leq 2\left(\frac{24\bar{a}B}{\Delta}\right)^M \exp\left(-\frac{cT_e\Delta^2}{16L^2}\right) =: C_M e^{-c'T_e}.$$

Because the supremum is over all $\mathbf{w}$, the bound does not require independence between $\widehat{\mathbf{W}}$ and unit i 's data. A union bound over units gives $\Pr(\hat{\gamma} \neq \gamma^0) \leq \Pr(E_\eta^c) + NC_M e^{-c'T_e}$, which tends to zero because $\log N/T_e \rightarrow 0$. This is (ii).

Step 4 (oracle equivalence). On the event $\{\hat{\gamma} = \gamma^0\}$, $\widehat{\mathbf{W}}$ minimises $\hat{R}(\gamma^0, \cdot)$, so $\widehat{\mathbf{W}} = \widetilde{\mathbf{W}}$ (if the minimiser is not unique, the two estimators select from the same set). By (ii) this event has probability approaching one. $\square$

B Out-of-sample risk

The theory concerns the population risk R evaluated at the estimated policy. In a rolling application the policy estimated on periods $1, \dots, T$ is applied in periods $T+1, \dots, T+k$. If $\{\mathbf{z}_{it}\}$ is β -mixing with coefficients $\beta(\cdot)$, a coupling argument (Yu, 1994) shows that the expected cost of the estimated policy in period $T+s$ differs from $\mathbb{E}R(\hat{\gamma}, \widehat{\mathbf{W}})$ by at most $L\beta(s)$. The bounds of Section 4 therefore also bound the out-of-sample cost up to a term that vanishes with the gap between estimation and application.

C Balancing authorities

D Additional results for the global panel

Table 16: The 46 balancing authorities, ordered by mean hourly demand over 2019–2026. Flex.: interchange flexibility in 2019 (range between the 5th and 95th percentiles of hourly total interchange, divided by mean demand). τ_i : calibrated critical ratio.

Code	Balancing authority	Mean MW	Flex.	τ_i
PJM	PJM Interconnection, LLC	92,501	0.06	0.947
MISO	Midcontinent Independent System Operator, Inc.	74,187	0.07	0.943
ERCO	Electric Reliability Council of Texas, Inc.	49,746	0.03	0.950
SWPP	Southwest Power Pool	32,259	0.08	0.940
SOCO	Southern Company Services, Inc. - Trans	27,072	0.12	0.927
CISO	California Independent System Operator	25,318	0.29	0.900
TVA	Tennessee Valley Authority	18,636	0.09	0.937
NYIS	New York Independent System Operator	17,305	0.18	0.910
FPL	Florida Power & Light Co.	15,870	0.10	0.933
ISNE	ISO New England	13,227	0.16	0.920
DUK	Duke Energy Carolinas	12,210	0.23	0.907
CPLE	Duke Energy Progress East	7,044	0.37	0.880
BPAT	Bonneville Power Administration	6,631	0.84	0.827
PACE	PacifiCorp East	5,803	0.42	0.877
PSCO	Public Service Company of Colorado	5,312	0.11	0.930
NEVP	Nevada Power Company	4,522	0.33	0.883
LGEE	Louisville Gas and Electric Company and Kentucky Utilities Company	4,166	0.17	0.913
AZPS	Arizona Public Service Company	3,893	0.67	0.843
SC	South Carolina Public Service Authority	2,901	0.32	0.890
LDWP	Los Angeles Department of Water and Power	2,861	0.30	0.897
SCEG	Dominion Energy South Carolina, Inc.	2,814	0.15	0.923
AECI	Associated Electric Cooperative, Inc.	2,658	0.52	0.867
TEC	Tampa Electric Company	2,530	0.26	0.903
PGE	Portland General Electric Company	2,523	0.53	0.863
PACW	PacifiCorp West	2,392	0.55	0.860
FMPP	Florida Municipal Power Pool	2,135	0.17	0.917
IPCO	Idaho Power Company	2,097	0.79	0.833
BANC	Balancing Authority of Northern California	1,993	0.59	0.853
PNM	Public Service Company of New Mexico	1,645	0.64	0.847
TEPC	Tucson Electric Power	1,590	0.71	0.837
JEA	JEA	1,532	0.32	0.893
AVA	Avista Corporation	1,460	0.70	0.840
NWMT	NorthWestern Corporation	1,358	1.07	0.810
SCL	Seattle City Light	1,076	0.92	0.817
WALC	Western Area Power Administration - Desert Southwest Region	1,045	0.95	0.813
EPE	El Paso Electric Company	1,029	0.46	0.873
GCPD	Public Utility District No. 2 of Grant County, Washington	696	1.58	0.807
CPLW	Duke Energy Progress West	564	0.61	0.850
TPWR	City of Tacoma, Department of Public Utilities, Light Division	540	0.56	0.857
IID	Imperial Irrigation District	441	0.91	0.820
TIDC	Turlock Irrigation District	328	0.81	0.830
TAL	City of Tallahassee	322	0.32	0.887
DOPD	PUD No. 1 of Douglas County	245	2.20	0.803
GVL	Gainesville Regional Utilities	240	0.50	0.870
CHPD	Public Utility District No. 1 of Chelan County	218	4.70	0.800
HST	City of Homestead	71	0.90	0.823

Table 17: Global panel without the weather candidate (eight candidates): system cost relative to the single best forecaster for estimation windows of 14 and 28 days (world).

Method	$W = 14$	$W = 28$
Single best forecaster ^a	1.000	1.000
Equal weights	1.315	1.315
Best single (per area)	0.887	0.870
Pooled DF weights	0.874	0.872
Per-area DF weights	0.857	0.838
Shrinkage to pooled	0.846	0.833
Two-step k -means	0.843	0.832
Forecast-then-commit, per area	0.973	0.923
Forecast-then-commit, grouped	0.948	0.917
Neural gate (deep MoE)	0.918	0.918
GDMA	0.849	0.835
Soft GDMA	0.841	0.828
Fair soft GDMA	—	—

Table 18: Global panel without the weather candidate and with a common critical ratio $\tau = 0.9$ for every grid area ($W = 28$): system cost relative to the single best forecaster. The neural gate and the fair variant were not run for this specification.

Method	World	North America	Europe	Oceania	Asia	Africa
Single best forecaster ^a	1.000	1.000	1.000	1.000	1.000	1.000
Equal weights	1.285	1.434	1.171	1.027	1.164	1.032
Best single (per area)	0.862	0.834	0.889	0.860	0.888	0.878
Pooled DF weights	0.865	0.834	0.893	0.920	0.864	0.828
Per-area DF weights	0.832	0.788	0.869	0.856	0.886	0.839
Shrinkage to pooled	0.828	0.787	0.863	0.865	0.857	0.817
Two-step k -means	0.826	0.793	0.853	0.858	0.857	0.837
Forecast-then-commit, per area	0.894	0.887	0.880	0.854	1.039	0.846
Forecast-then-commit, grouped	0.888	0.903	0.855	0.853	1.006	0.819
GDMA	0.828	0.796	0.854	0.868	0.859	0.812
Soft GDMA	0.822	0.786	0.853	0.859	0.857	0.810